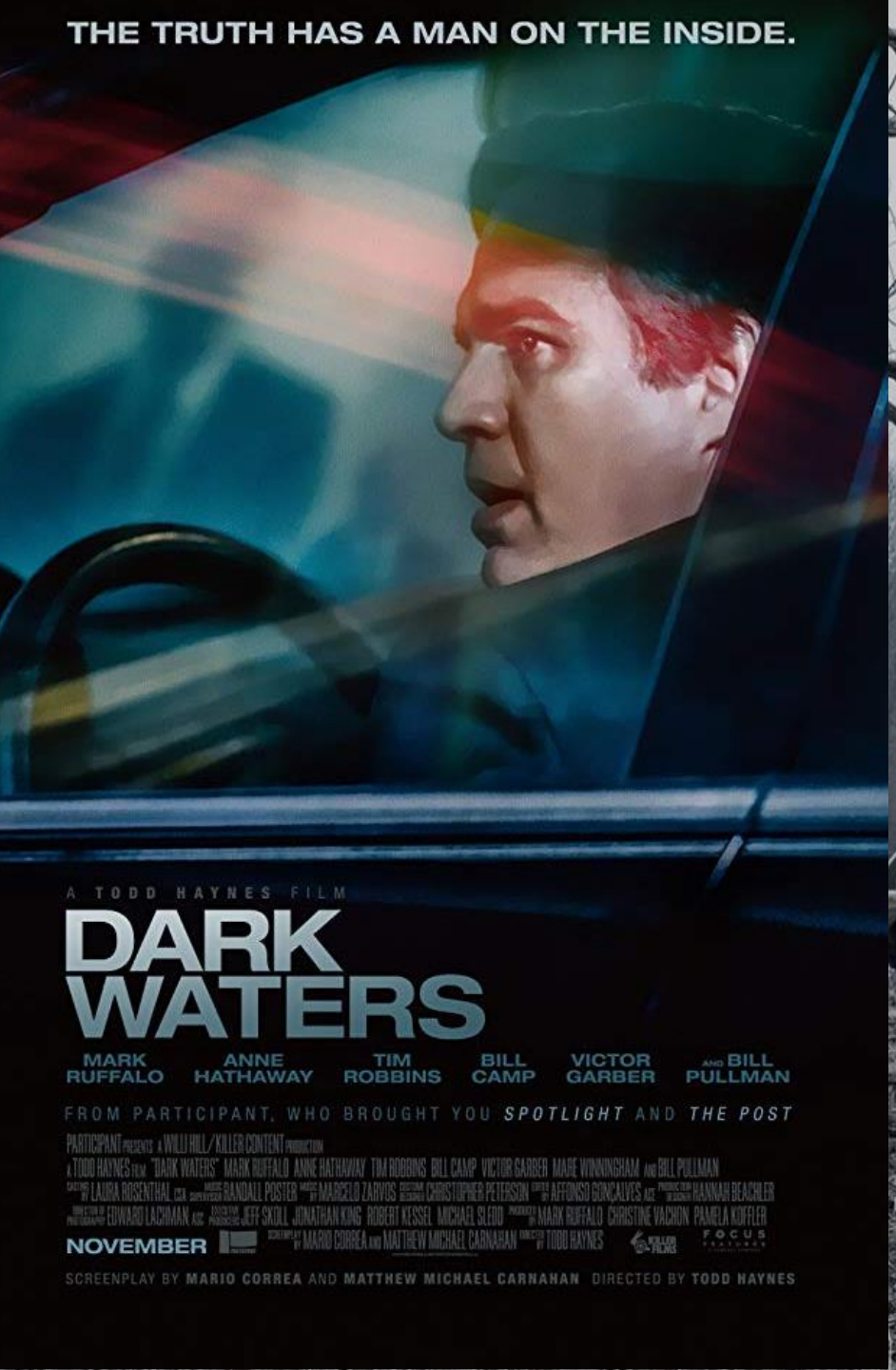
HWRS 561b: Physical Hydrogeology II

PFAS (Part 1)

Agenda

- Introduction to PFAS





The New York Times Magazine

FEATURE

January 2016

The Lawyer Who Became DuPont's Worst Nightmare

Rob Bilott was a corporate defense attorney for eight years. Then he took on an environmental suit that would upend his entire career — and expose a brazen, decades-long history of chemical pollution.

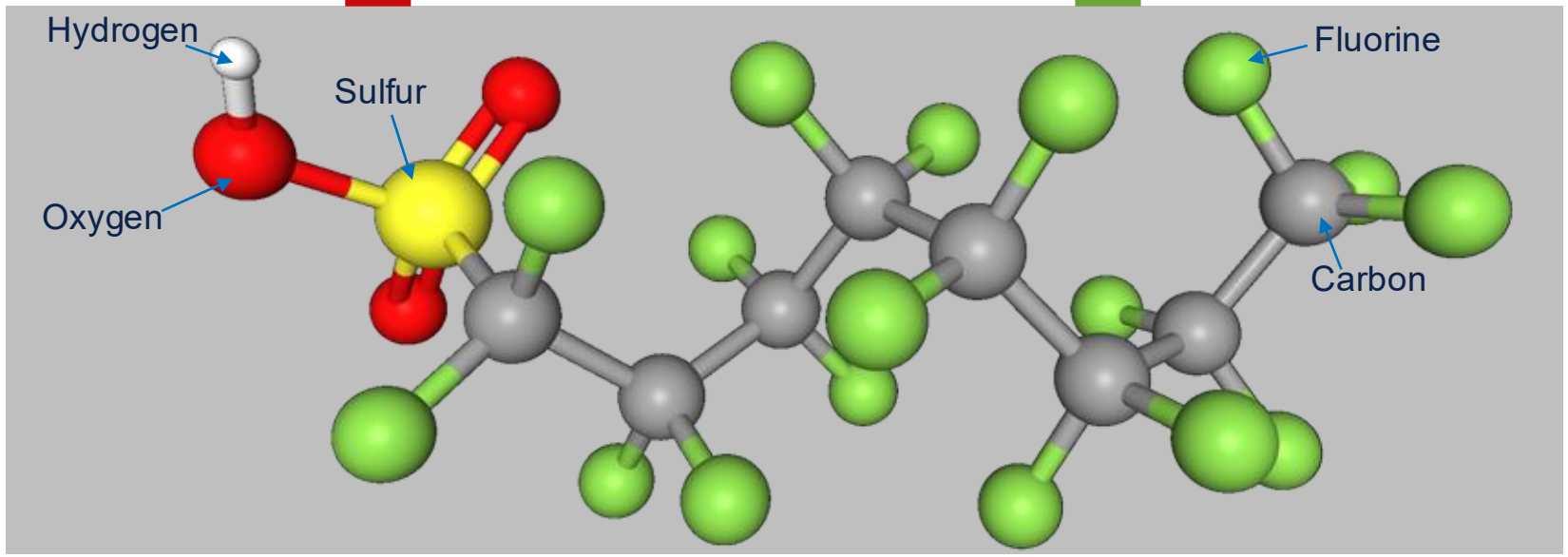
Based on a true story – Lawsuit against DuPont led to total settlement of **$753 million**

What are PFAS (Per- and poly-FluoroAlkyl Substances)?

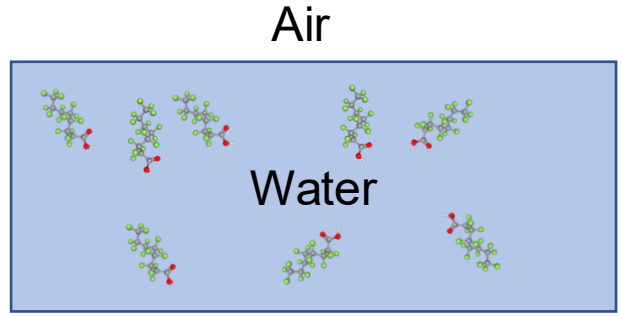
PFAS molecule (e.g., PFOS)

(“likes” water) Hydrophilic headgroup

(“does not like” water) Hydrophobic-oleophobic tail



- Surfactant (**Surf**ace **ac**tive ag**ent**)
- Persistent (**C-F** bond)
- Toxic at **ppt** levels
- More than **14,000** compounds



Accumulate at air–water interfaces

Used in our daily life and at military sites

"Perfect" chemicals if NOT toxic

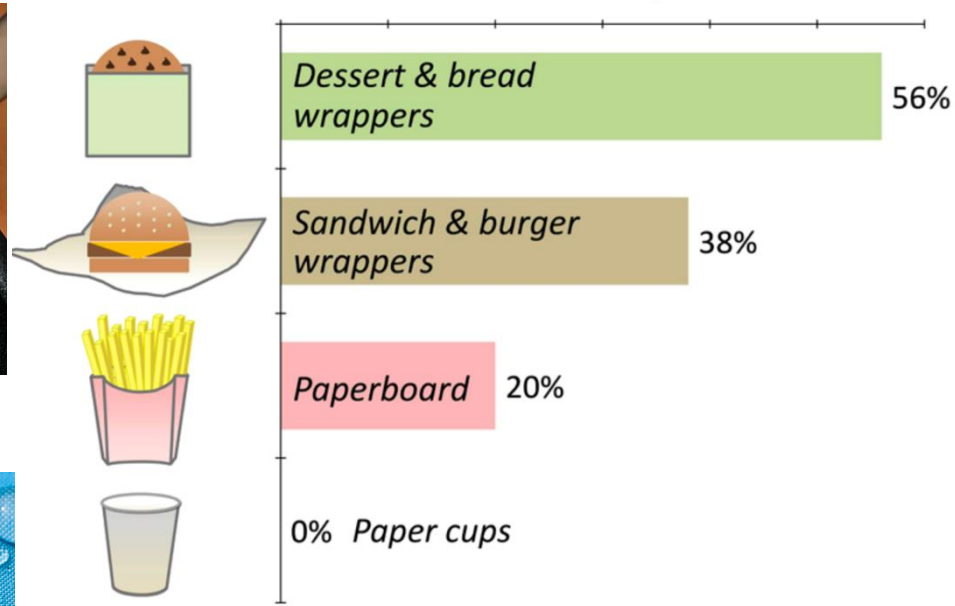
Non-stick, stain- and water-resistant coating





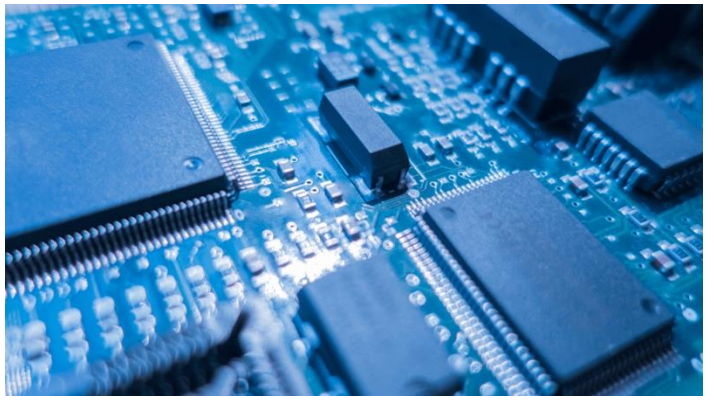
Food packaging

Percent with fluorine



Schaider et al (2017)

Semiconductor Industry

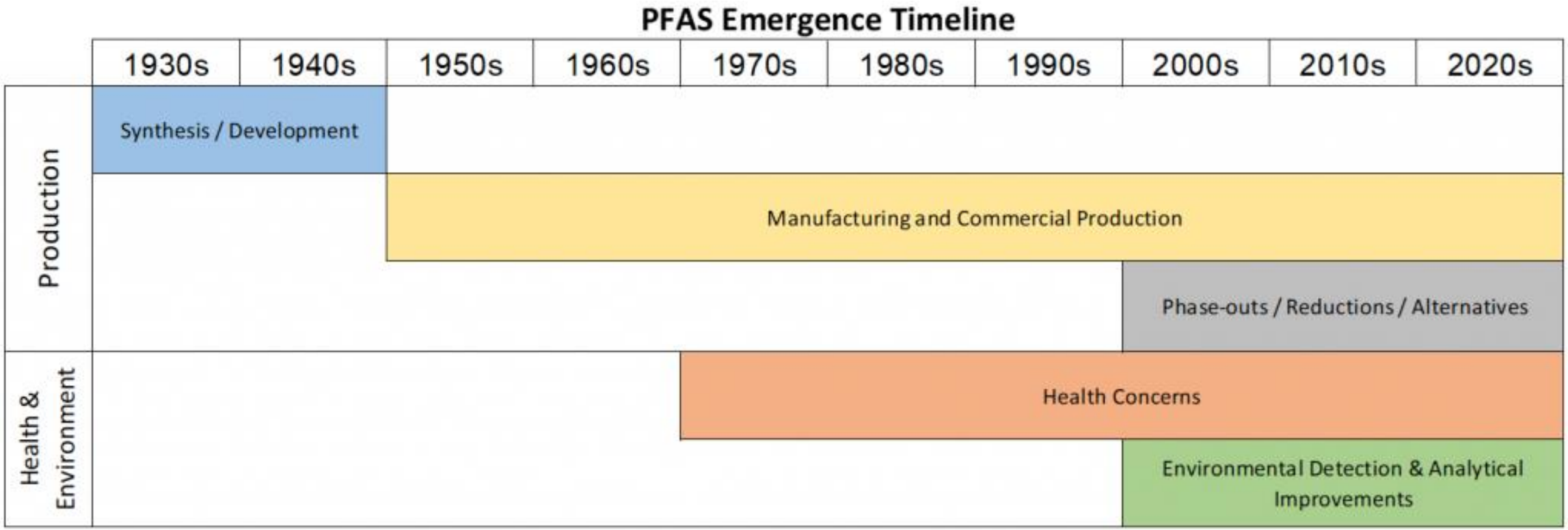


Fire fighting foam



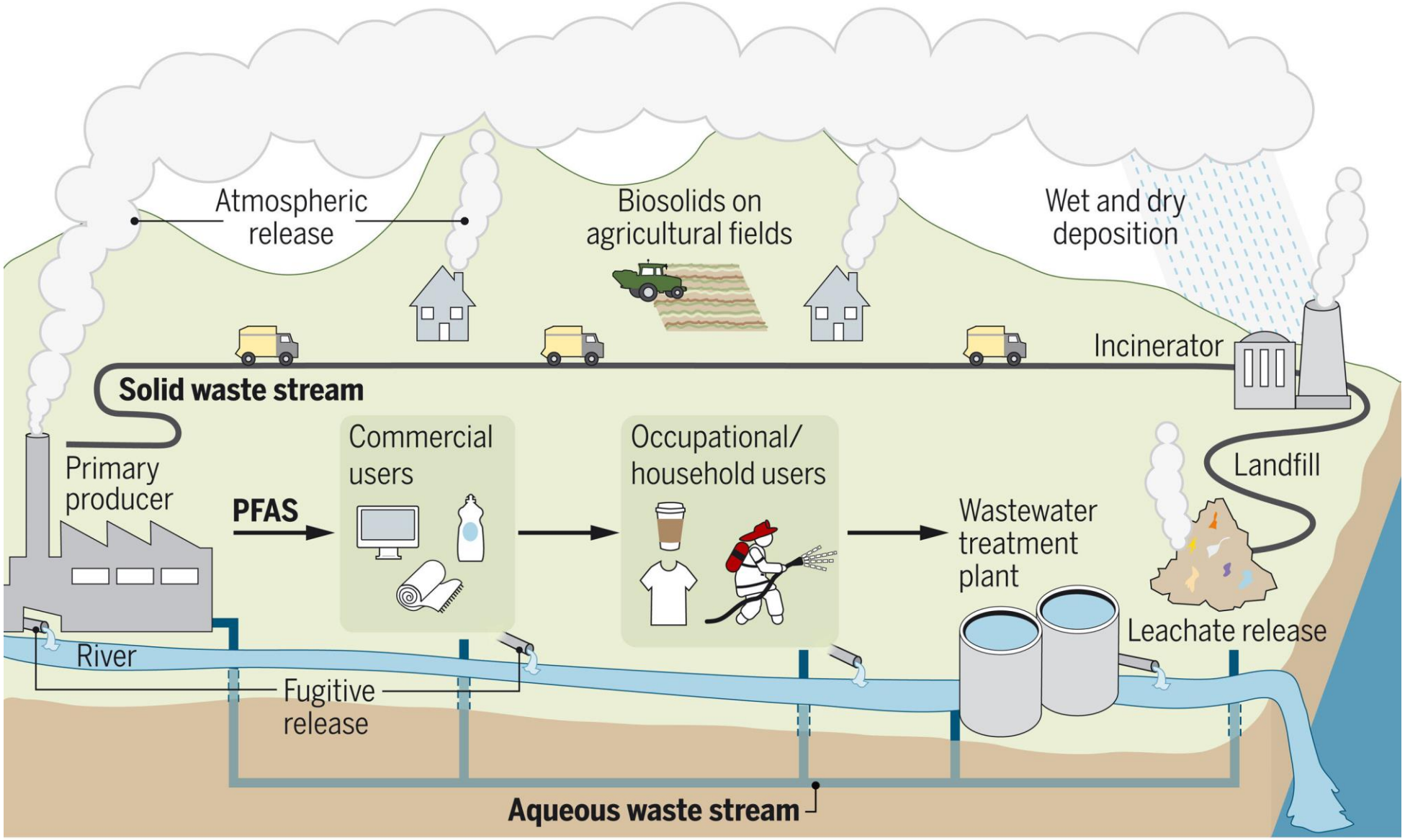
http://www.safetynews.co.nz/fire-fighting-foams-causing-sparks-fly/

General timeline of PFAS emergence and awareness



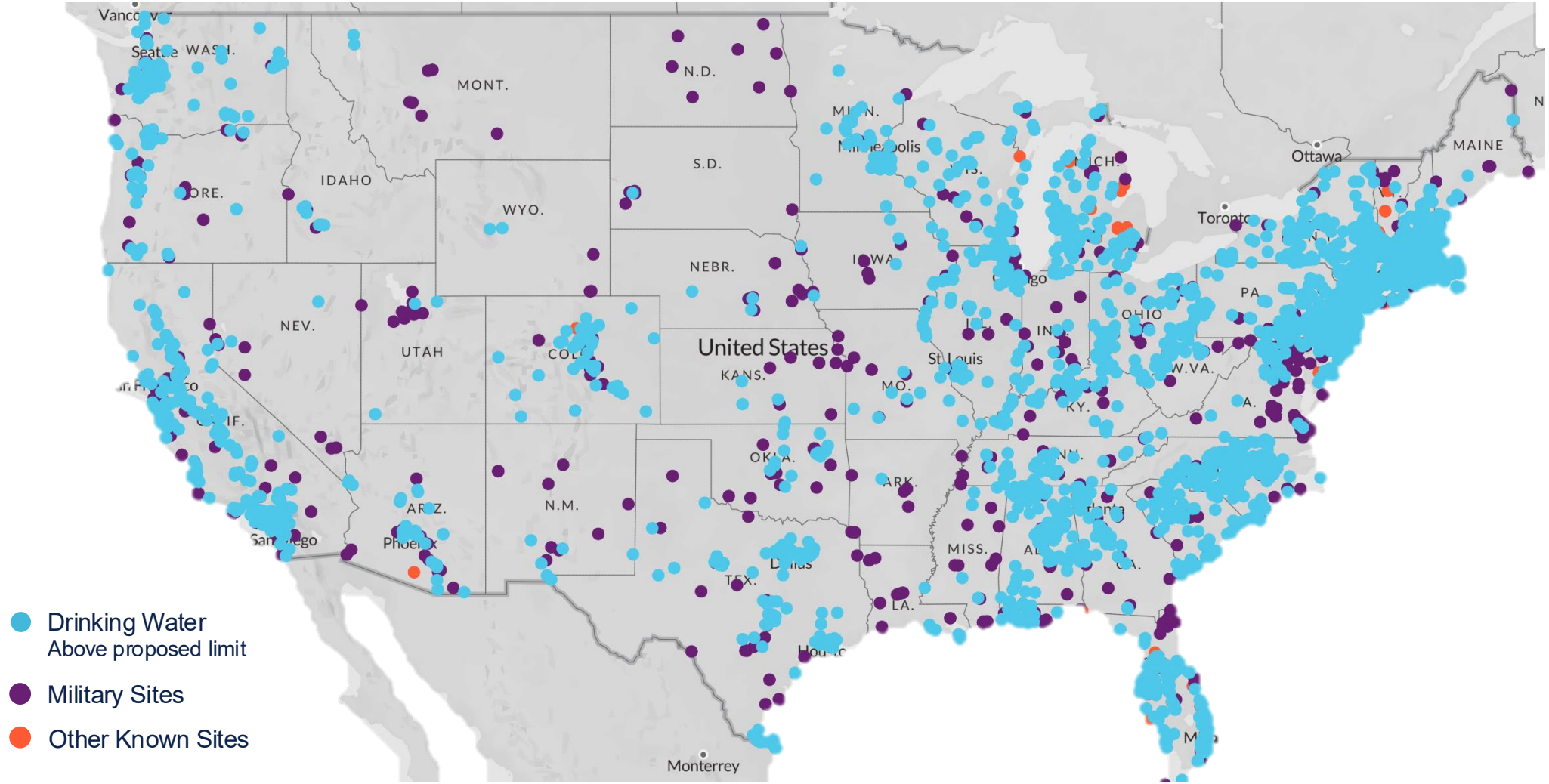
Source: J. Hale, Kleinfelder and ITRC PFAS fact sheets

PFAS releases to the environment



Source: Evich et al., 2022. Science.

PFAS are widely spread in the environment

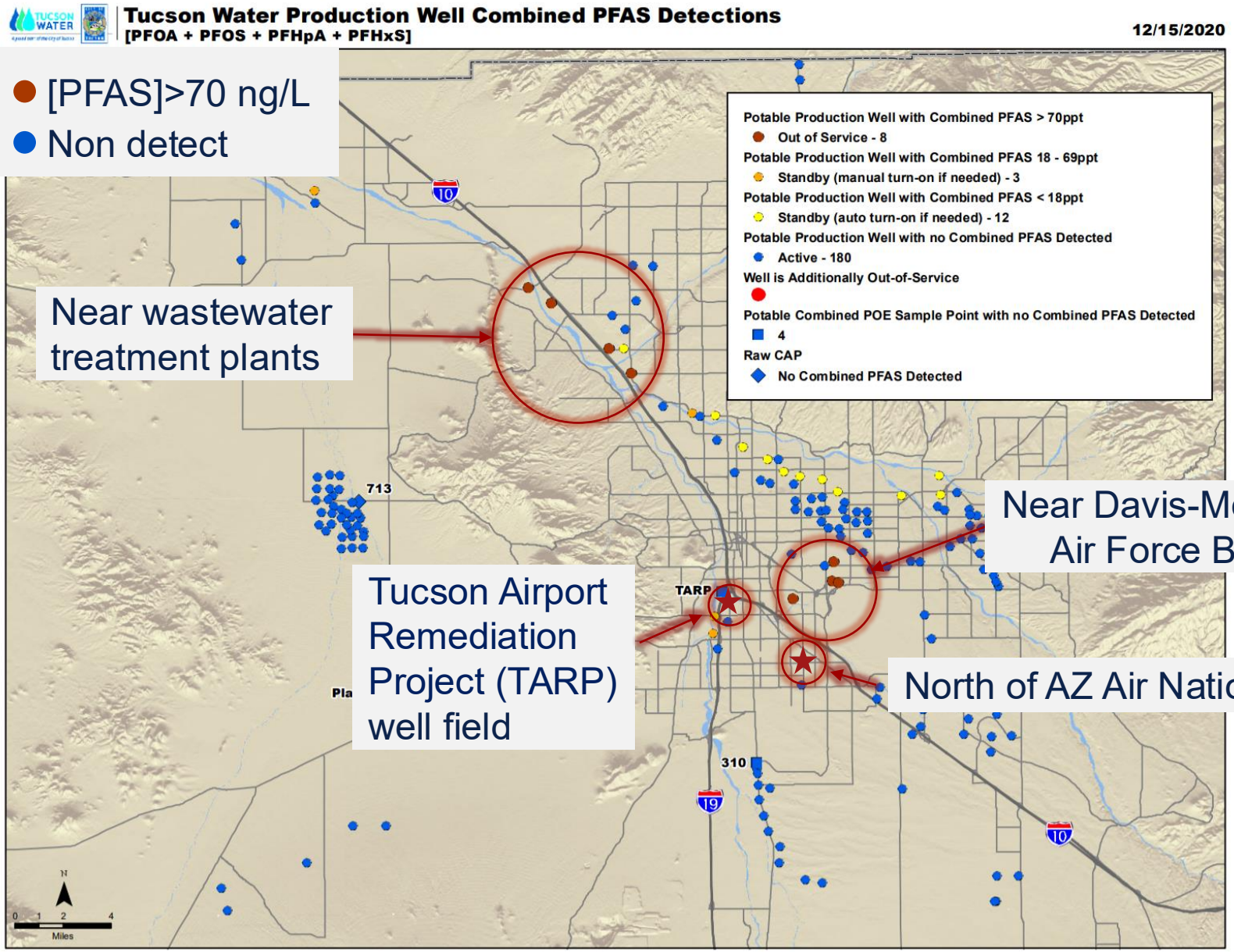


3500+ reported PFAS contamination sites in the US.

https://www.ewg.org/interactive-maps/pfas_contamination/map/

PFAS contamination in Tucson, AZ

- [PFAS]>70 ng/L
- Non detect



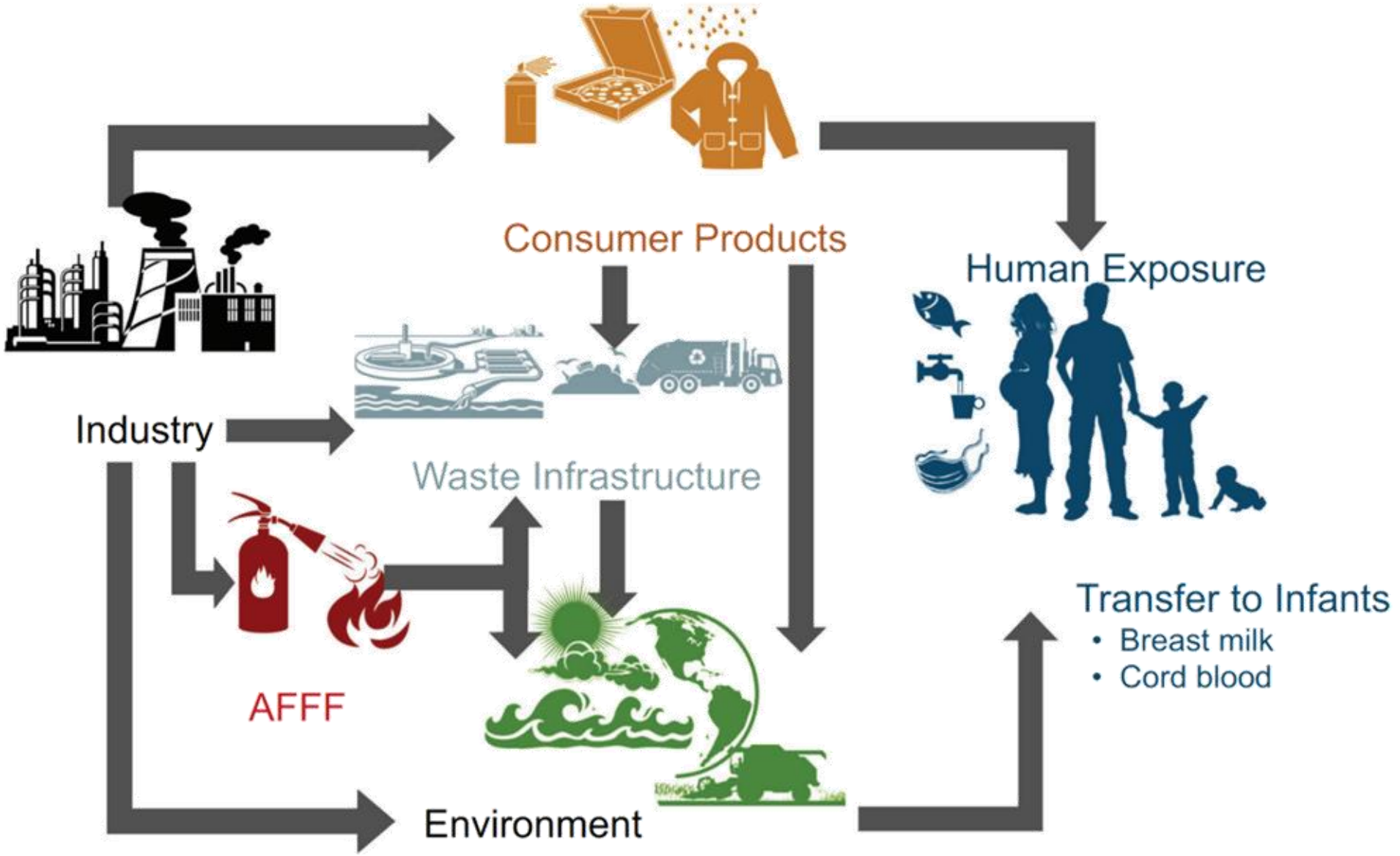
Near wastewater treatment plants

Tucson Airport Remediation Project (TARP) well field

Near Davis-Monthan Air Force Base

North of AZ Air National Guard

97% of US residents have PFAS in their blood



Sunderland et al (2019)

What we know about health effects of PFAS exposure?

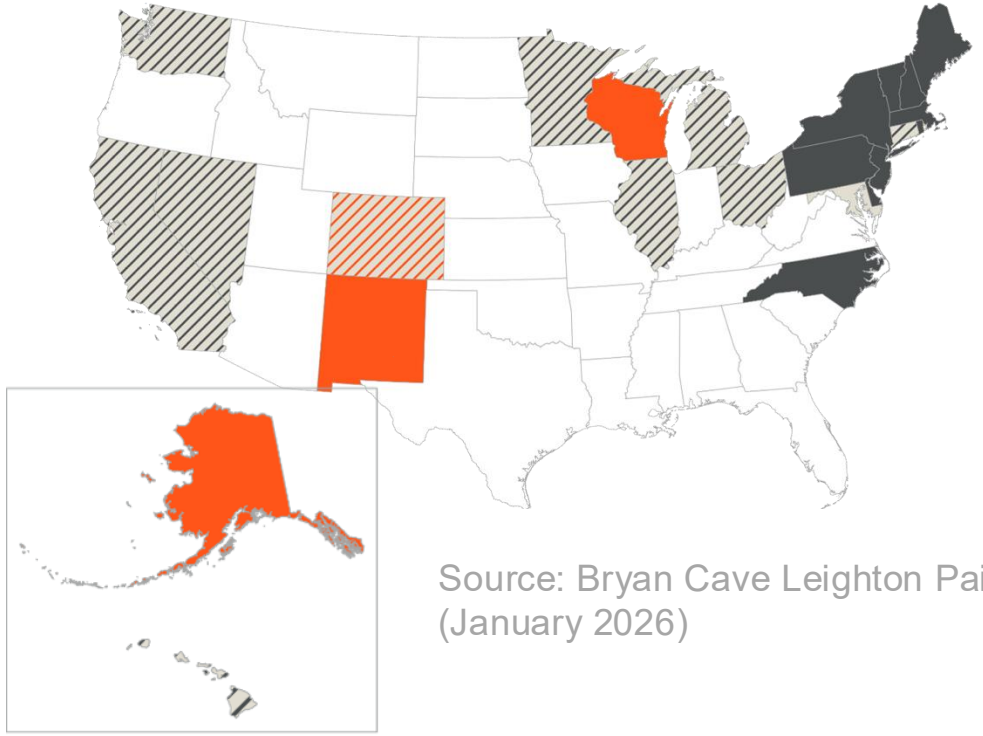
- **According to EPA, exposure to certain levels of PFAS may lead to:**
  - *Reproductive effects* such as decreased fertility or increased high blood pressure in pregnant women.
  - *Developmental effects* or delays in children, including low birth weight, accelerated puberty, bone variations, or behavioral changes.
  - Increased risk of some ***cancers***, including prostate, kidney, and testicular cancers.
  - Reduced ability of the body's *immune system* to fight infections, including reduced vaccine response.
  - Interference with the body's natural *hormones*.
  - *Increased cholesterol* levels and/or risk of *obesity*.

- Residents in Parkersburg, West Virginia affected by PFAS contamination have higher rates of various health problems including cancers (Robert Bilott's lawsuit against DuPont. ***Movie: Dark Waters***)

  "In 2004, DuPont settled Bilott's class-action suit, which now applied to 80,000 plaintifs in 6 water districts, for $343 million. The settlement also included the creation of a C8 Science Panel and funding of a study to collect medical information on the exposed population and determine whether PFOA exposure actually posed harm. The project was complete in 2013, and linked PFOA exposure to 6 diseases: ulcerative colitis, pregnancy-induced hypertension, thyroid disease, testicular cancer, and kidney cancer (Lerner, 2015)."

https://www.epa.gov/pfas/our-current-understanding-human-health-and-environmental-risks-pfas
https://pfasproject.com/parkersburg-west-virginia/

Regulation of PFAS in drinking water in the US

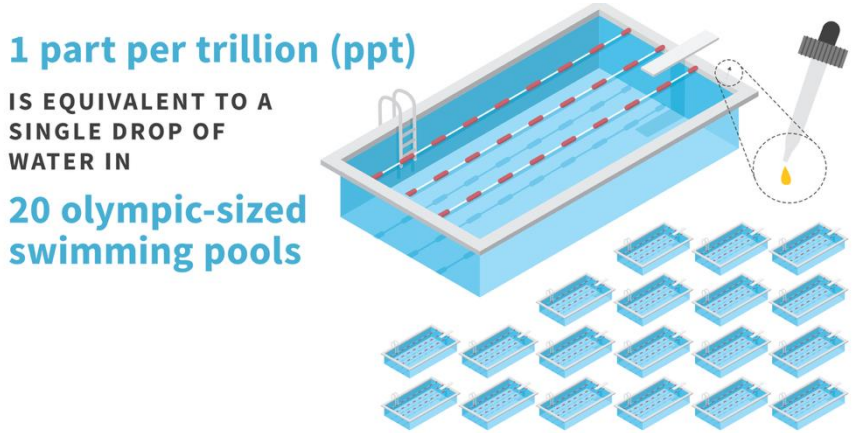


- PFAS not regulated in drinking water
- Regulated (> 70ppt)
- Regulated (=70 ppt)
- Regulated (<70 ppt)
- Multiple standards (some above and some below 70 ppt)
- Multiple standards (some equal to and some above 70 ppt)

Source: Bryan Cave Leighton Paisner (January 2026)

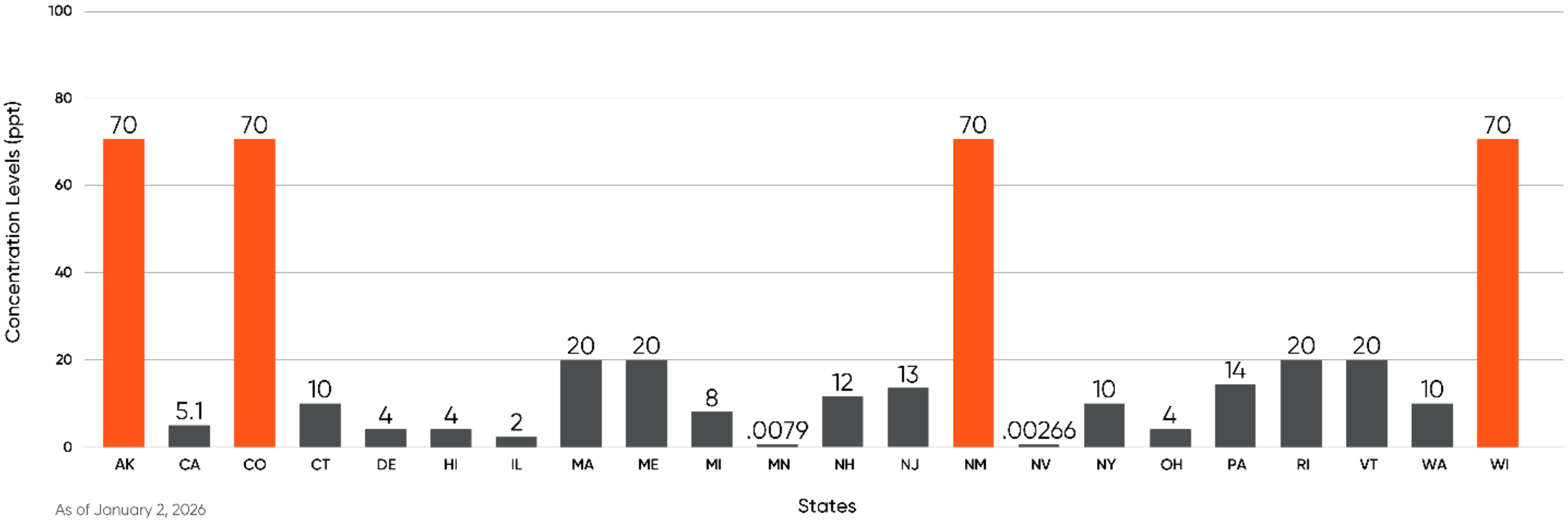
- Not regulated yet by EPA, but States are aggressively setting their own regulatory standards
- EPA proposed unprecedented MCLs (March 14, 2023)
  - PFOA = 4 parts per trillion (ppt or ng/L)
  - PFOS = 4 parts per trillion (ppt or ng/L)
  - PFNA, PFHxS, PFBS, HFPO-DA (GenX): Hazard Index = 1

$$\text{Hazard Index} = \frac{[\text{PFNA}]}{10\,\text{ppt}} + \frac{[\text{PFHxS}]}{9\,\text{ppt}} + \frac{[\text{PFBS}]}{2000\,\text{ppt}} + \frac{[\text{GenX}]}{10\,\text{ppt}}$$



https://www.michigan.gov/

State regulatory limits for PFOA and/or PFOS



Source: Bryan Cave Leighton Paisner (January 2026)

How do the proposed MCLs compare to other contaminants?

Regulatory criteria for contaminant group (Newell et al., 2020)

Groundwater contaminant	Regulatory criteria (µg/L)	Source
Many CVOCs such as TCE, PCE	5	Maximum concentration limit (MCL) for many CVOCs
1,4-Dioxane	0.46	USEPA drinking water screening level
Benzene	5	MCL for benzene
MTBE	13	California maximum contaminant limit for drinking water (2017)
PFOA/PFOS	0.004	EPA proposed MCLs

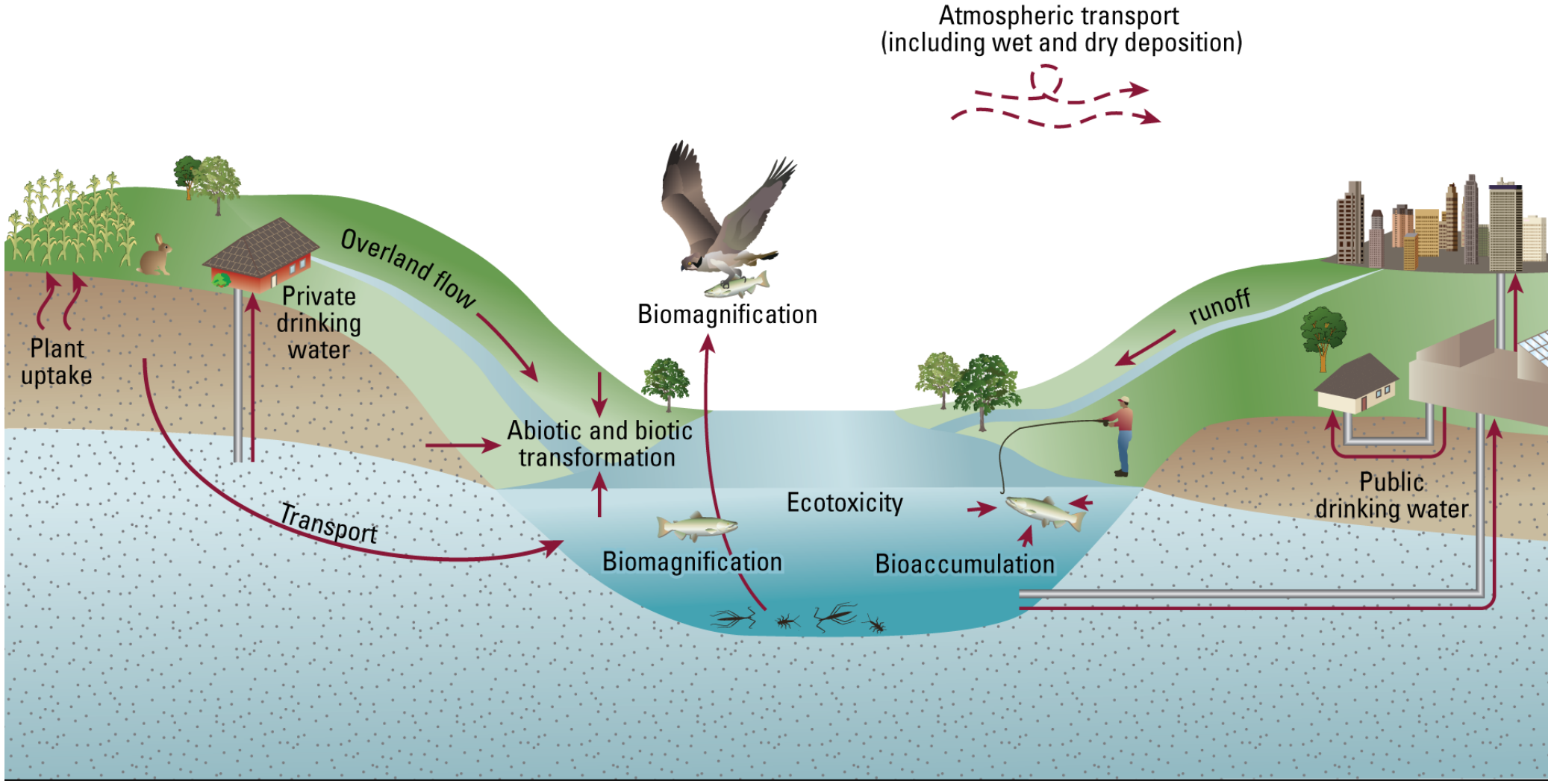
Tucson Airport Remediation Project (TARP) was created 1994 to treat TCE, and then later 1,4-Dioxane.

- The MCLs for PFAS are 100 to 1000 times lower than many other groundwater contaminants.

Abbreviations: CVOC, chlorinated volatile organic compound; MTBE, Methyl tert-butyl ether; PCE, perchloroethene; PFOA, perfluorooctanoic acid; PFOS, perfluorosulfonic acid; USEPA, U.S. Environmental Protection Agency.

How can hydrogeology help solve the PFAS problem?

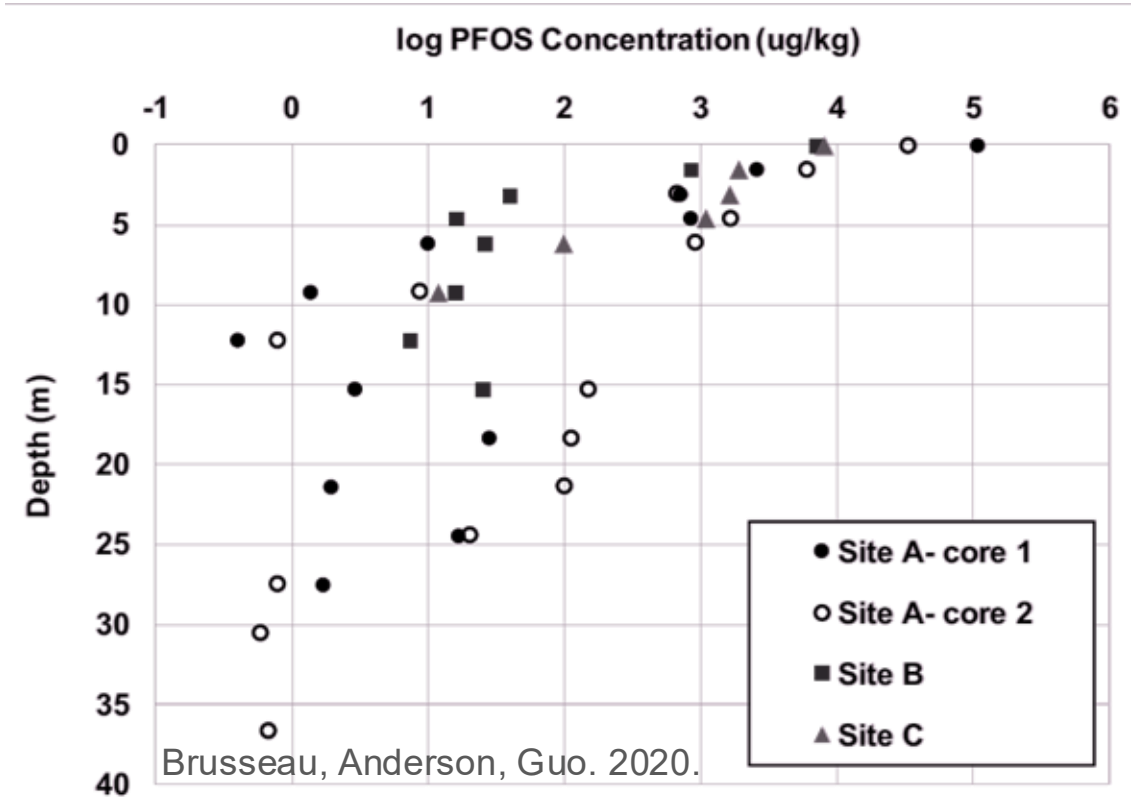
PFAS fate and transport in the environment



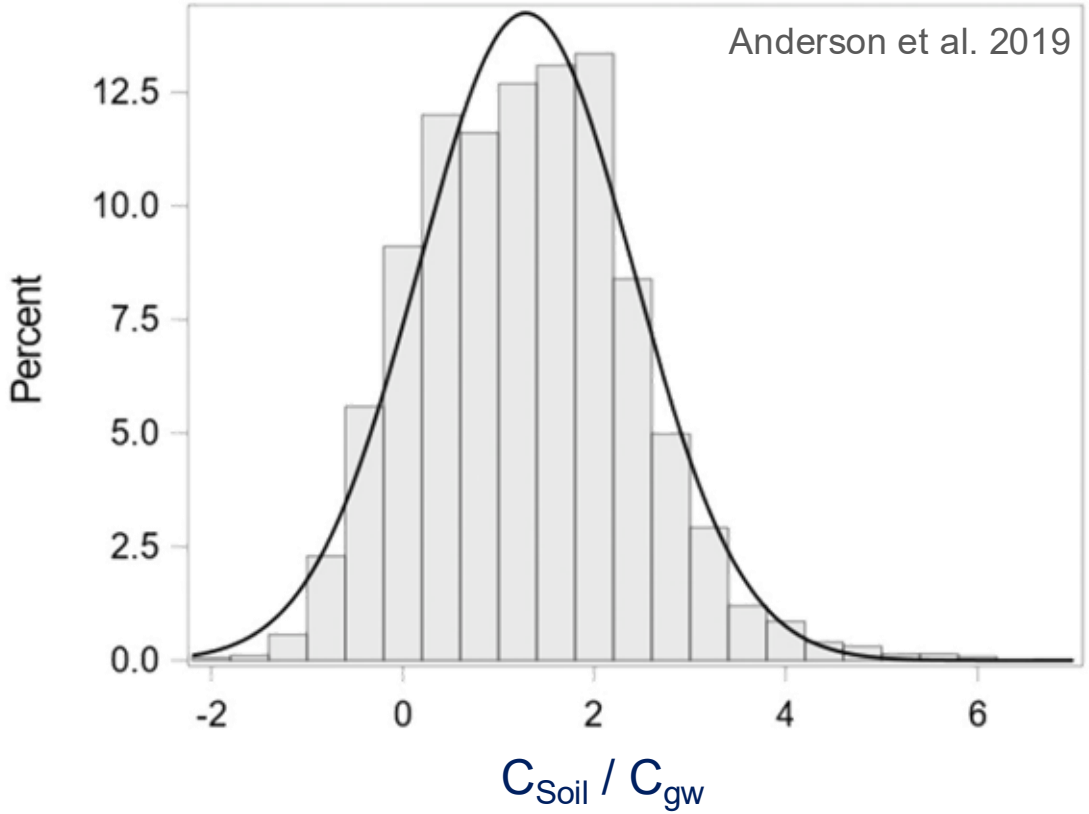
Source: USGS

- Water cycle is a major driver of PFAS movement in the environment.

Field data: concentrations in soil vs. groundwater



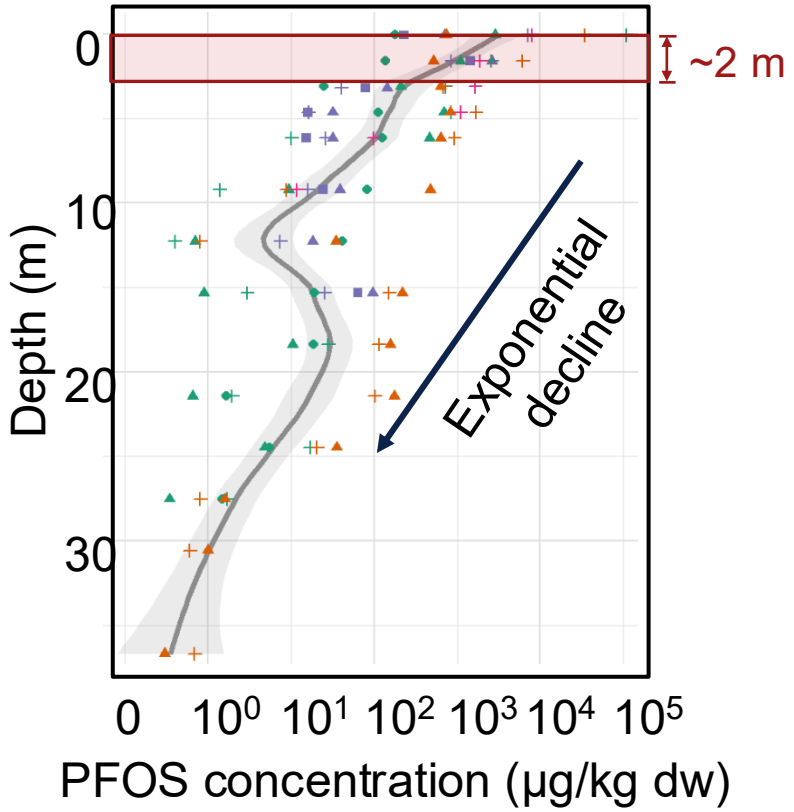
PFOS concentration in soil

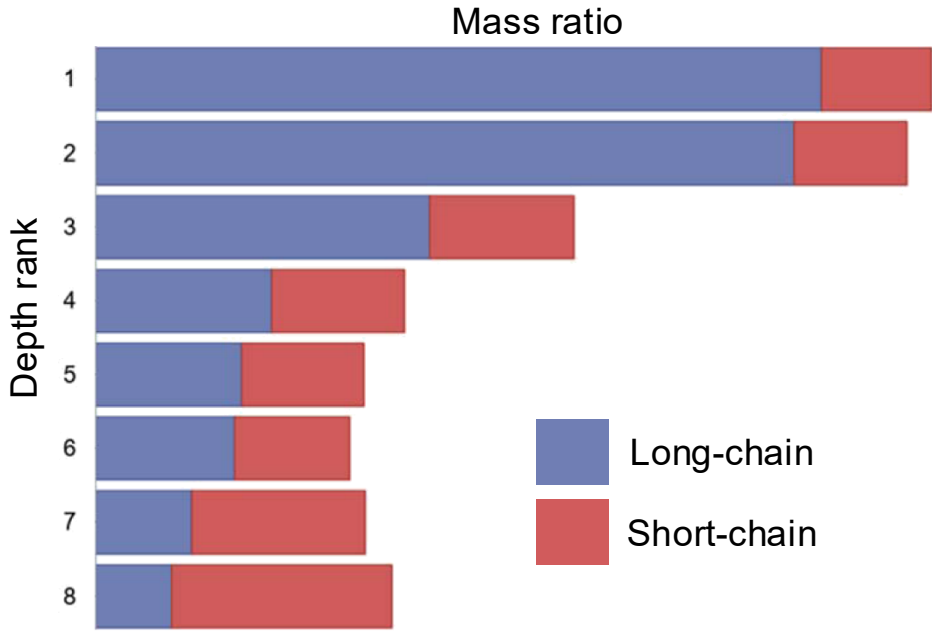


Ratio between soil and groundwater PFAS concentrations (324 sites across 56 military installations throughout the continental U.S.)

- Concentrations in soils are generally much greater than that in groundwater.

Field data: spatial variation of concentrations in the soil





- Soils appear to act as significant source zones of PFAS.
- Long-chain PFAS tend to retain in shallow soil, while short-chain PFAS migrate to deeper depth.

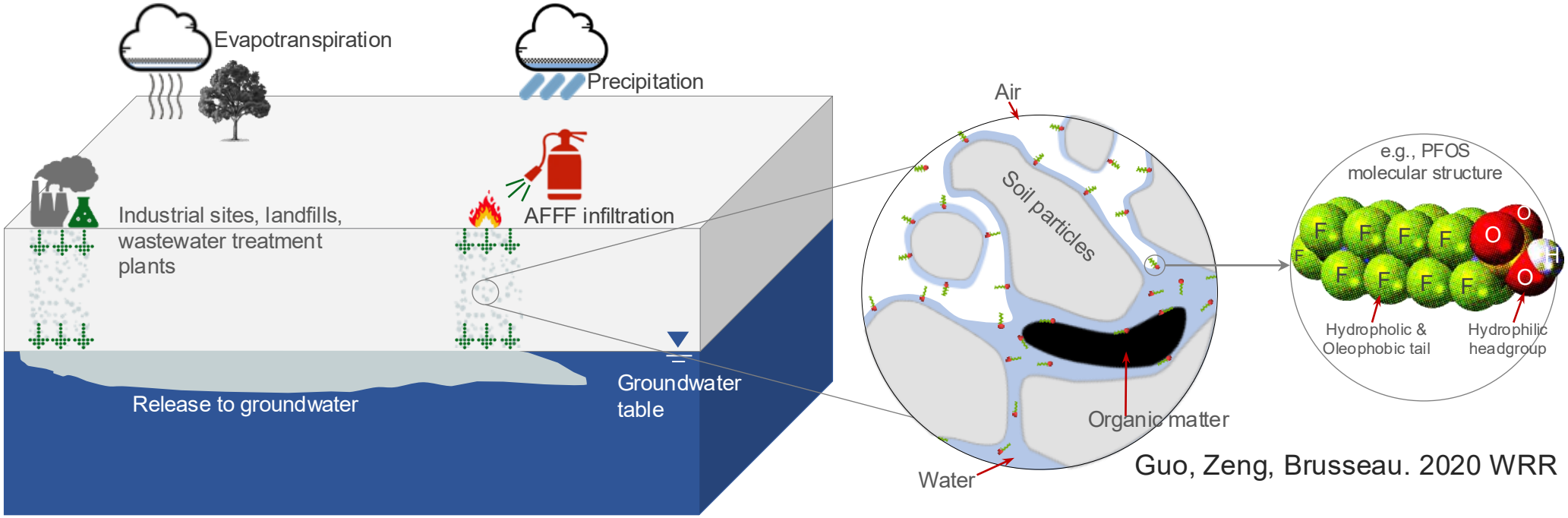
Anderson et al. 2019
Brusseau, Anderson, Guo. 2020.

Overarching Questions

What are the primary processes controlling PFAS leaching in soils?

How can we quantify the long-term mass discharge rates to groundwater?

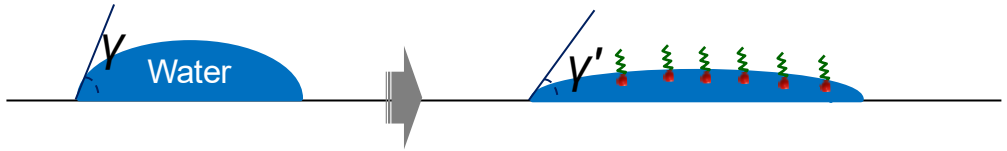
PFAS transport in soils: physical & chemical processes



Guo, Zeng, Brusseau. 2020 WRR

Unique physicochemical properties:

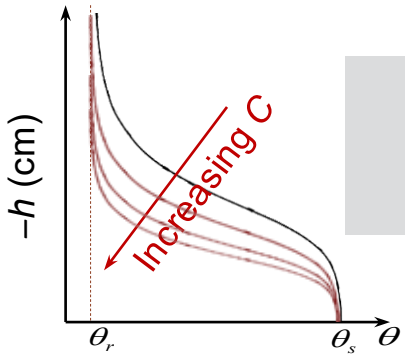
- ✓ PFAS accumulate at solid surfaces and air-water interfaces in soils.
- ✓ PFAS present in pore water can modify surface tension and contact angle.



PFAS transport in soils: mathematical formulation

- Soil water flow:

$$\frac{\partial \theta}{\partial t} - \nabla \cdot [\boldsymbol{K}(\theta)\nabla(h - z)] = 0$$



Surfactant-induced flow

$$h = \frac{\sigma(C)\cos\gamma(C)}{\sigma_0 \cos\gamma_0} h_0$$

(Leverett J scaling)

- PFAS transport:

$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \boldsymbol{v} C) - \nabla \cdot (\theta \boldsymbol{D} \nabla C) = 0$$

Advection Dispersion

Nonlinear and rate-limited solid-phase adsorption

$$C_{s,1} = F_s K_f C^N \qquad \frac{dC_{s,2}}{dt} = \alpha_s[(1 - F_s)K_f C^N - C_{s,2}]$$

Instantaneous Kinetic

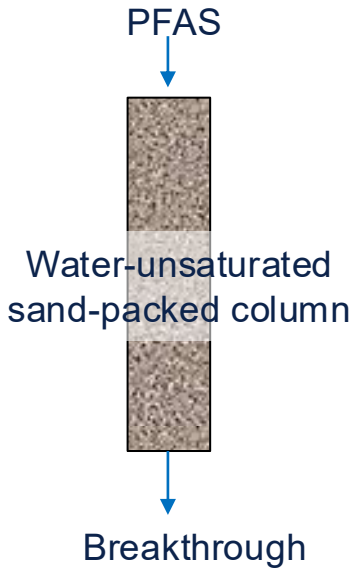
Nonlinear and rate-limited air-water interfacial adsorption

$$C_{aw,1} = F_{aw} A_{aw} K_{aw} C \qquad \frac{dC_{aw,2}}{dt} = \alpha_{aw}[(1 - F_{aw})A_{aw}K_{aw}C - C_{aw,2}]$$

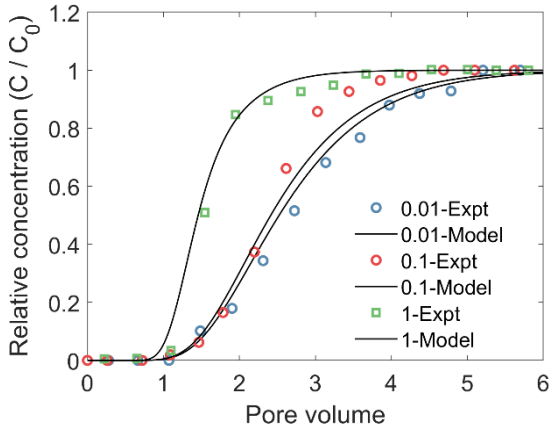
Instantaneous Kinetic

Guo, Zeng, Brusseau. 2020 WRR
Zeng & Guo. 2021 AWR

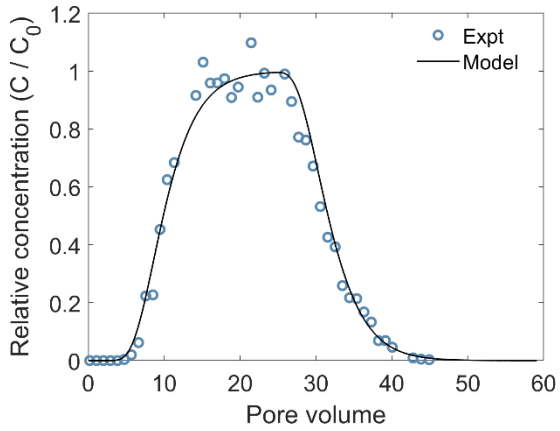
Model validation: vs. miscible-displacement experiments



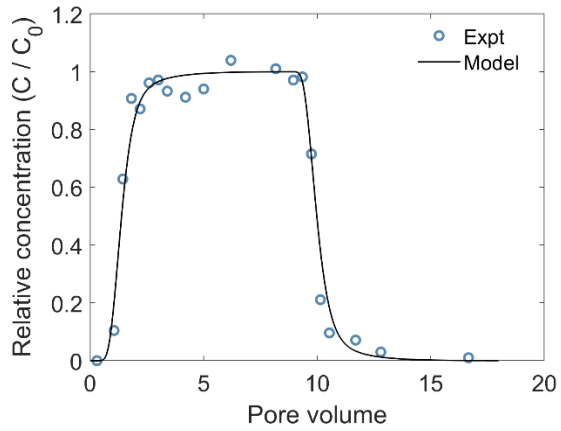
Experimental data set #1
PFOA (1 mg/L, 0.1 mg/L, 0.01 mg/L)
Measured data: Lyu et al., 2018



Experimental data set #2
PFOS (0.1 mg/L)
Measured data: Brusseau et al., 2021



Experimental data set #3
GenX (10 mg/L)
Measured data: Yan et al., 2021

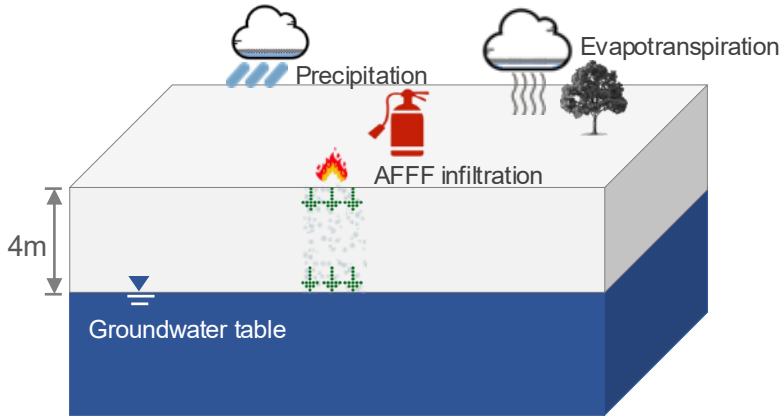


- Independent model predictions match well with experimental data.

Zeng, Brusseau, Guo. 2021 JH
Guo, Zeng, Brusseau, Zhang. 2022 AWR

Numerical simulation: PFAS migration at a fire training area

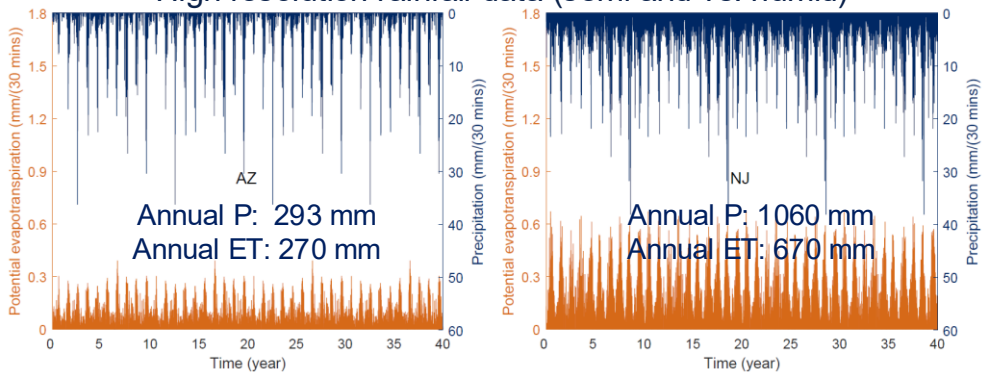
- PFAS contamination scenario at an AFFF-impacted fire training area site



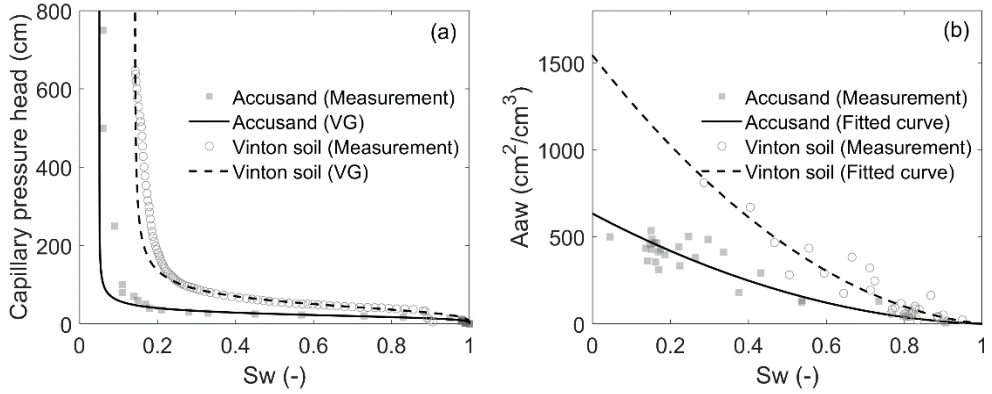
- ✓ Fire training: one session every 10 days lasting for 30 years
- ✓ Representative PFAS mixture in 1% diluted AFFF solution

  PFOS: 100 mg/L, PFHxS: 7.1 mg/L, PFBS: 1.4 mg/L

High resolution rainfall data (semi-arid vs. humid)

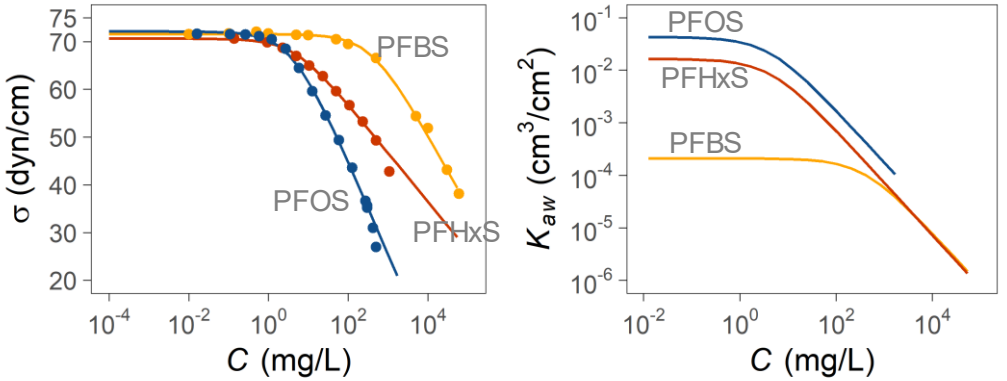


Two soil types (Accusand vs. Vinton)



Soil water characteristics | Air-water interfacial area

Three representative PFAS

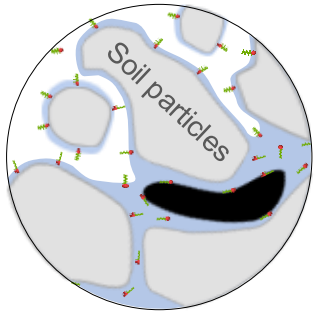


Surface tension data | Air-water interfacial adsorption coeff.

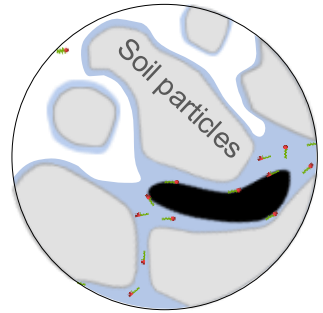
Numerical simulation: PFAS migration at a fire training area

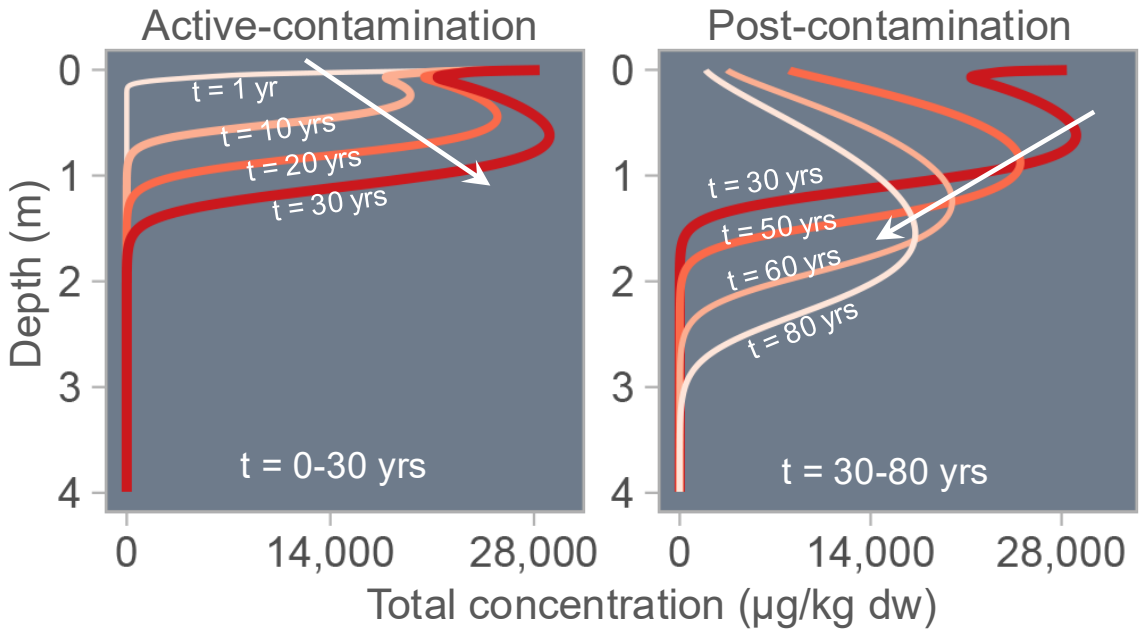
- Retention and leaching of PFOS: temporal evolution of spatial profiles

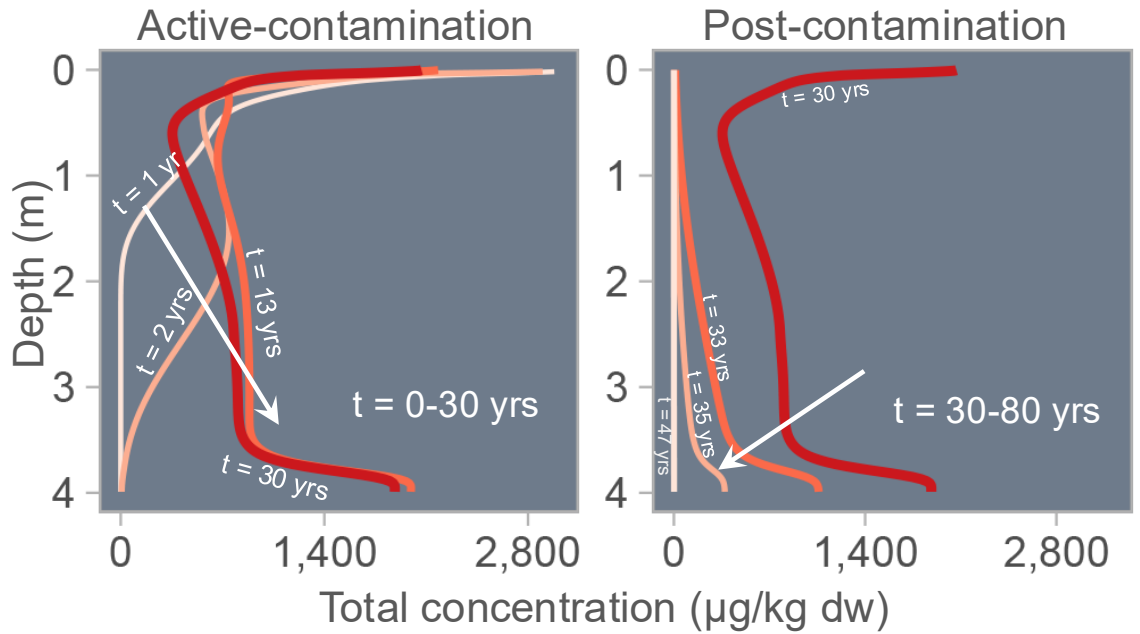
w/ AWI adsorption



w/o AWI adsorption





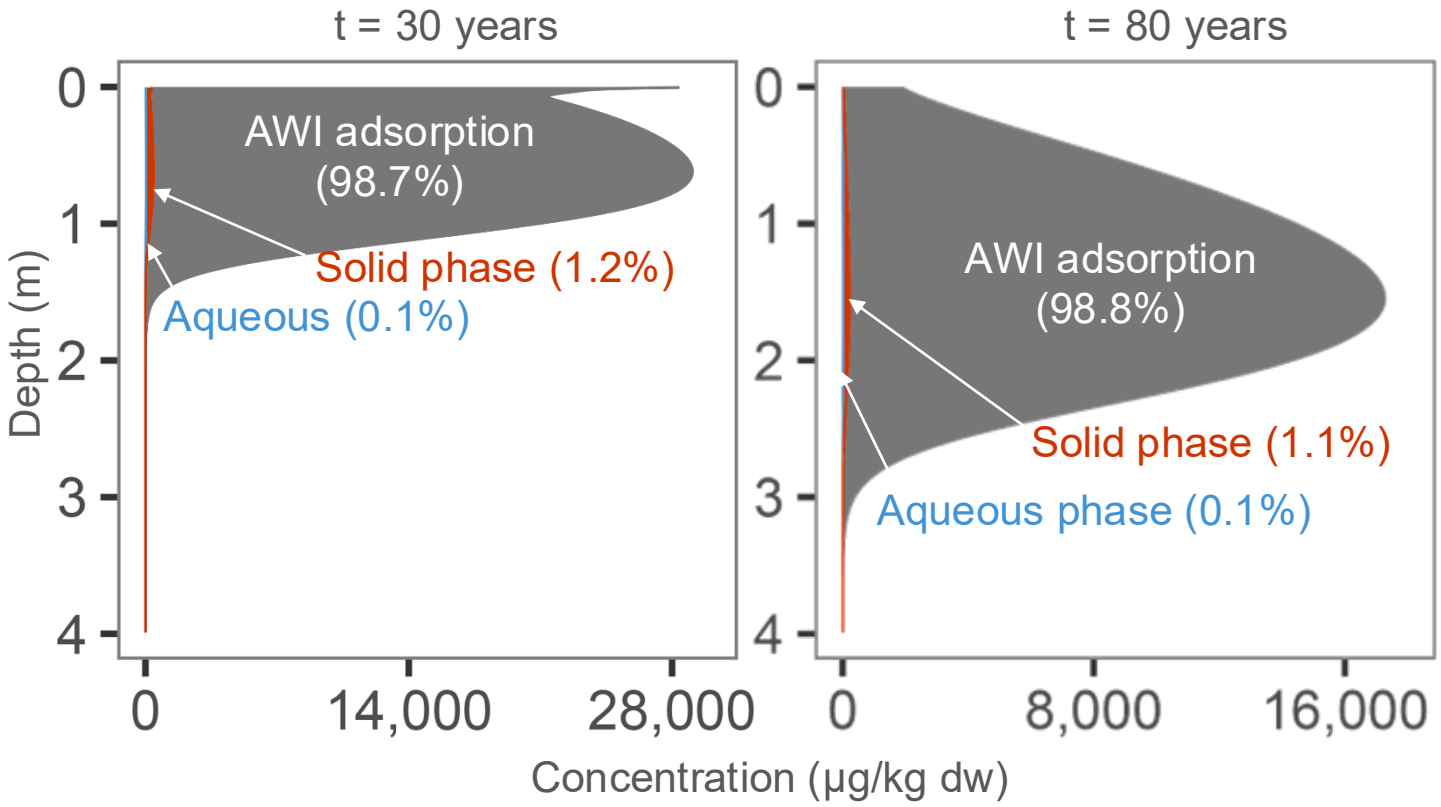


➢ Air-water interfacial adsorption significantly reduces the PFOS leaching in soils.

Guo, Zeng, Brusseau. 2020 WRR

Numerical simulation: PFAS migration at a fire training area

- Retention and leaching of PFOS: mass distribution in soils

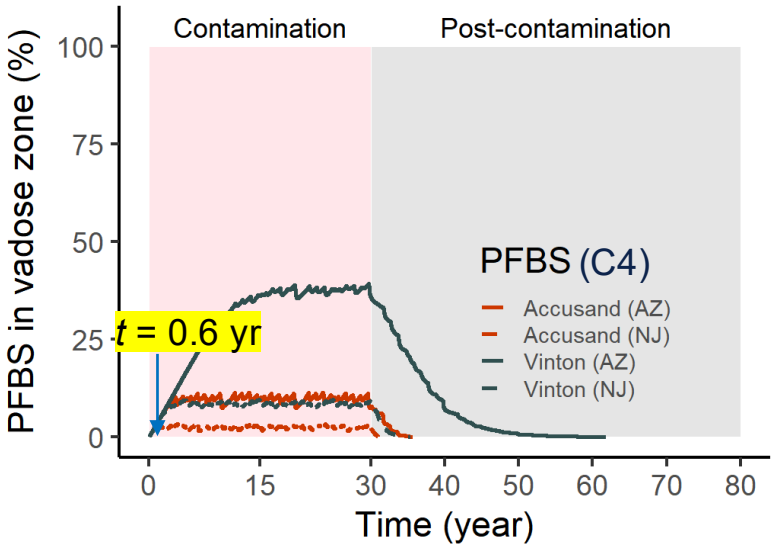


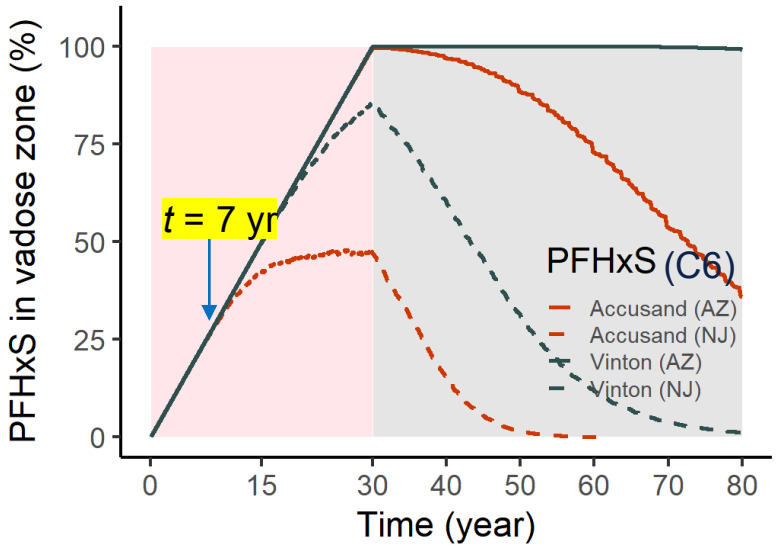
- The majority (>98%) of PFOS in the soil is adsorbed at the air-water interfaces.
- Only 0.1% and ~1% of PFOS in aqueous and solid phase.
- C in soils >> C in groundwater.

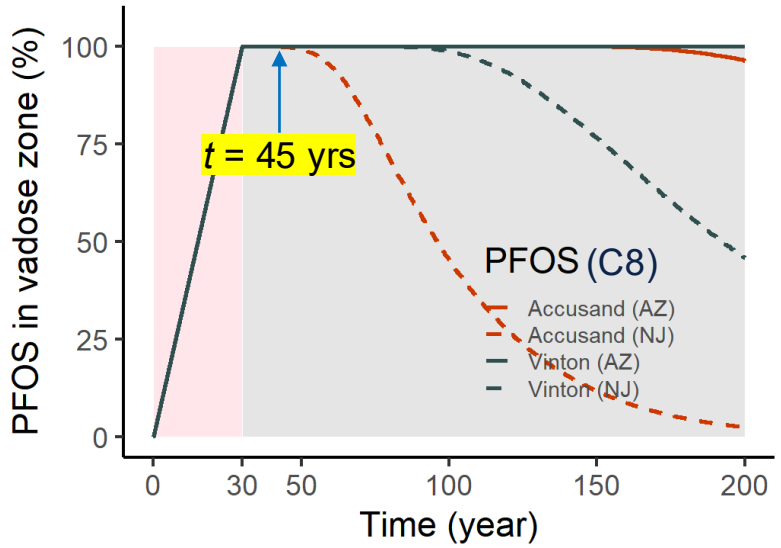
Guo, Zeng, Brusseau. 2020 WRR

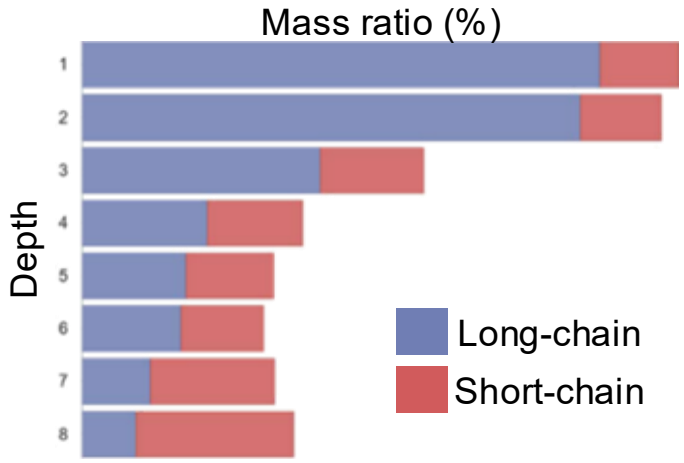
Numerical simulation: PFAS migration at a fire training area

- Short-chain vs. long-chain







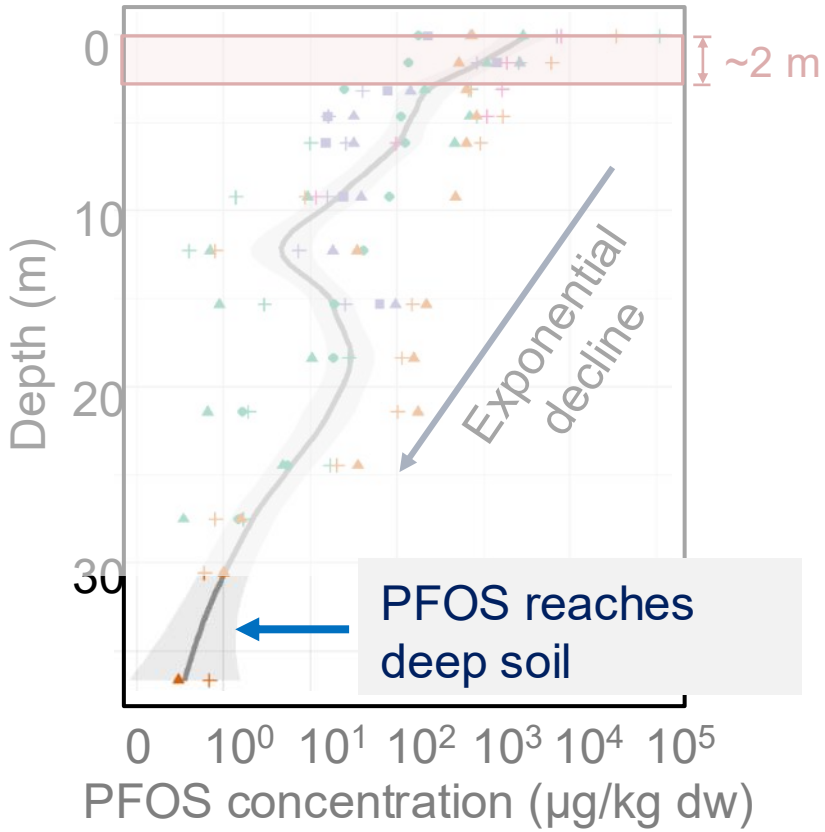


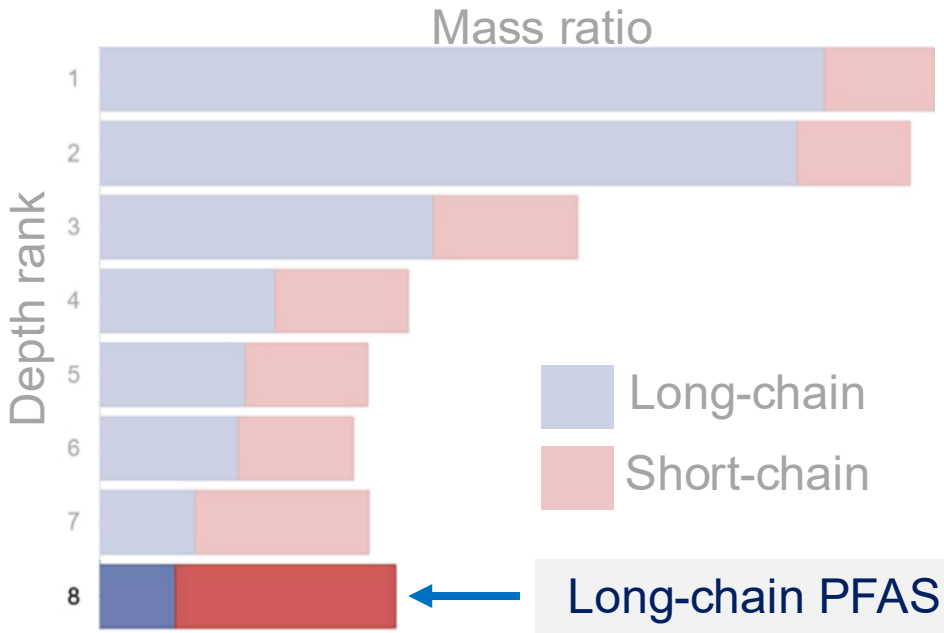
- PFBS, PFHxS, and PFOS reach groundwater table at t = 0.6 yrs, 7 yrs, and 45 yrs.
- PFOS is much more strongly retained in the soil than PFBS and PFHxS.
- Long-chain PFAS is retained in the shallow soil; while short-chain PFAS reach much deeper depth.

Zeng, Brusseau & Guo. 2021 JH

Field data: Spatial Profiles of PFAS in Vadose Zones

PFAS concentration profiles in the soil



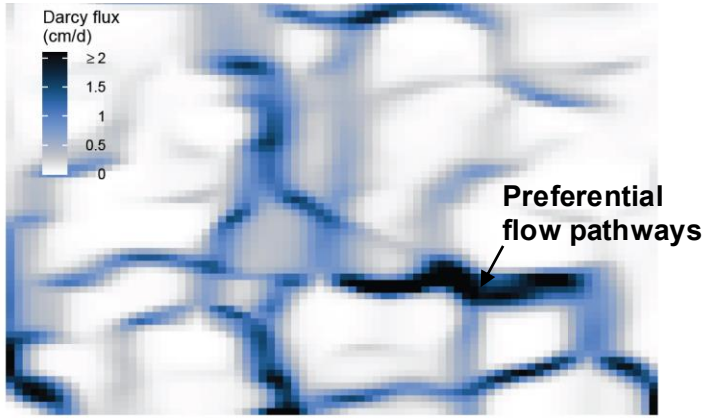


Data sources:
Anderson et al. 2019
Brusseau, Anderson, Guo. 2020.

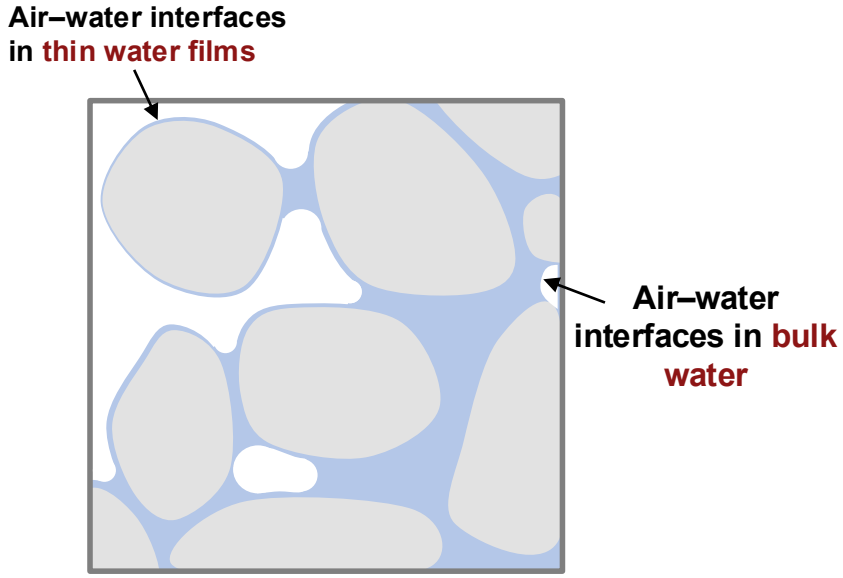
- The simulations capture strong retention of long-chain PFAS in shallow soil
- But they fail to represent leaching to deep soil and early arrival to groundwater

Hypotheses

- Heterogeneity-generated preferential flow leads to long-chain PFAS leaching to deep soil.

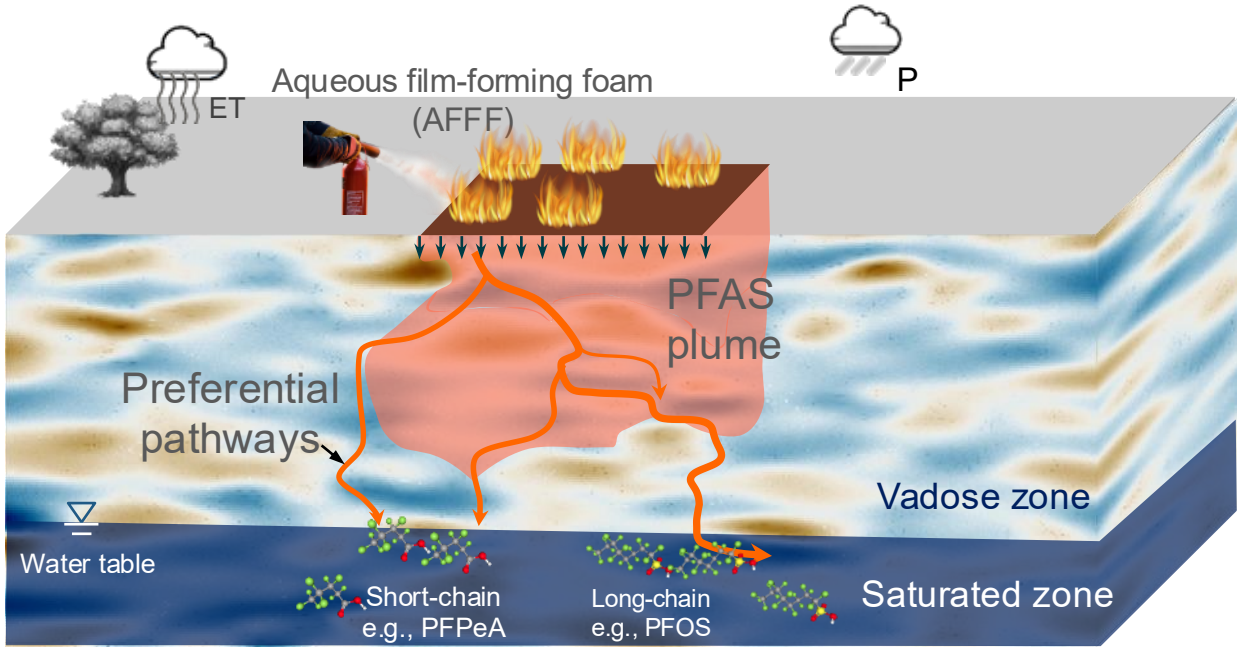


- Air-water interfacial area may not be fully accessible at the pore-scale.



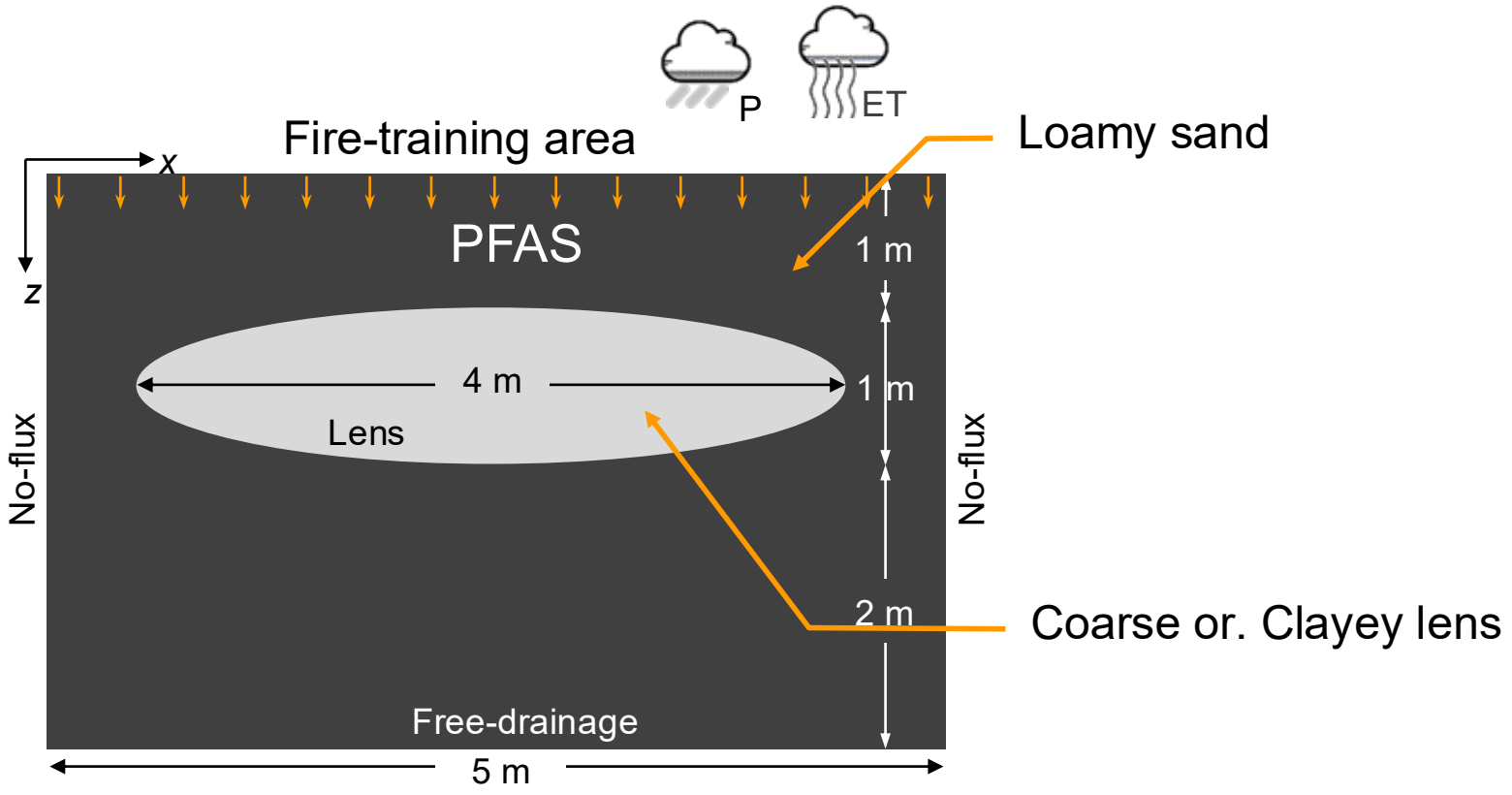
PFAS contamination at a fire-training area

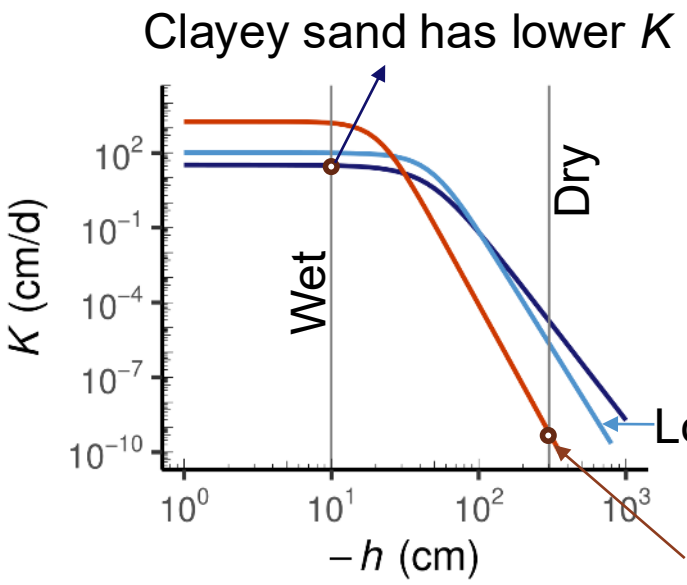
- Area: 30 m × 30 m.
- 1% diluted AFFF.
- Fire training: 30 min per 10 days for 30 yrs.
- Water table is deeper than 4 m.



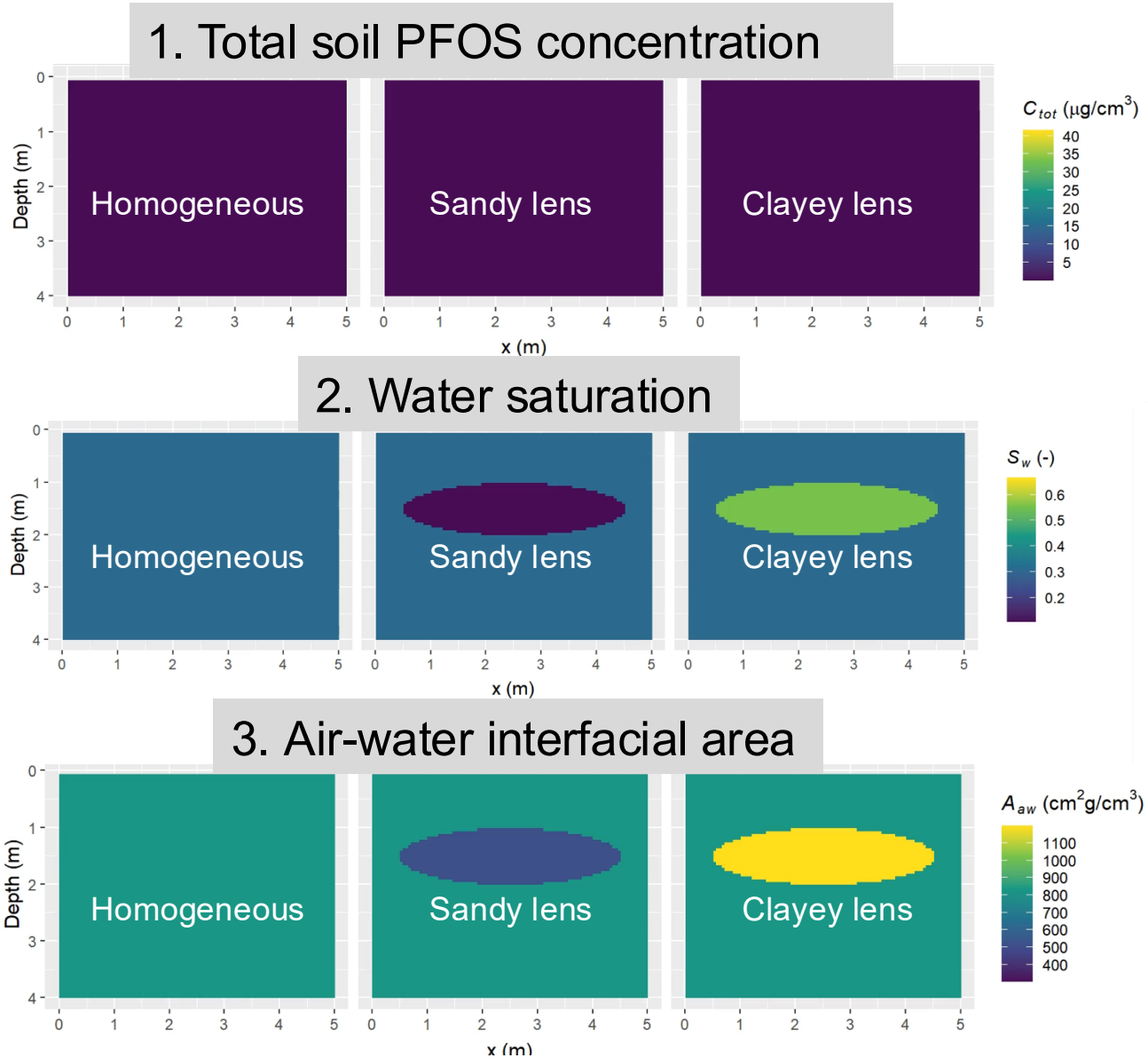
(Zeng, Guo, 2021; Guo, Zeng, Brusseau, 2020)

Soil lenses





Zeng & Guo. 2021 AWR

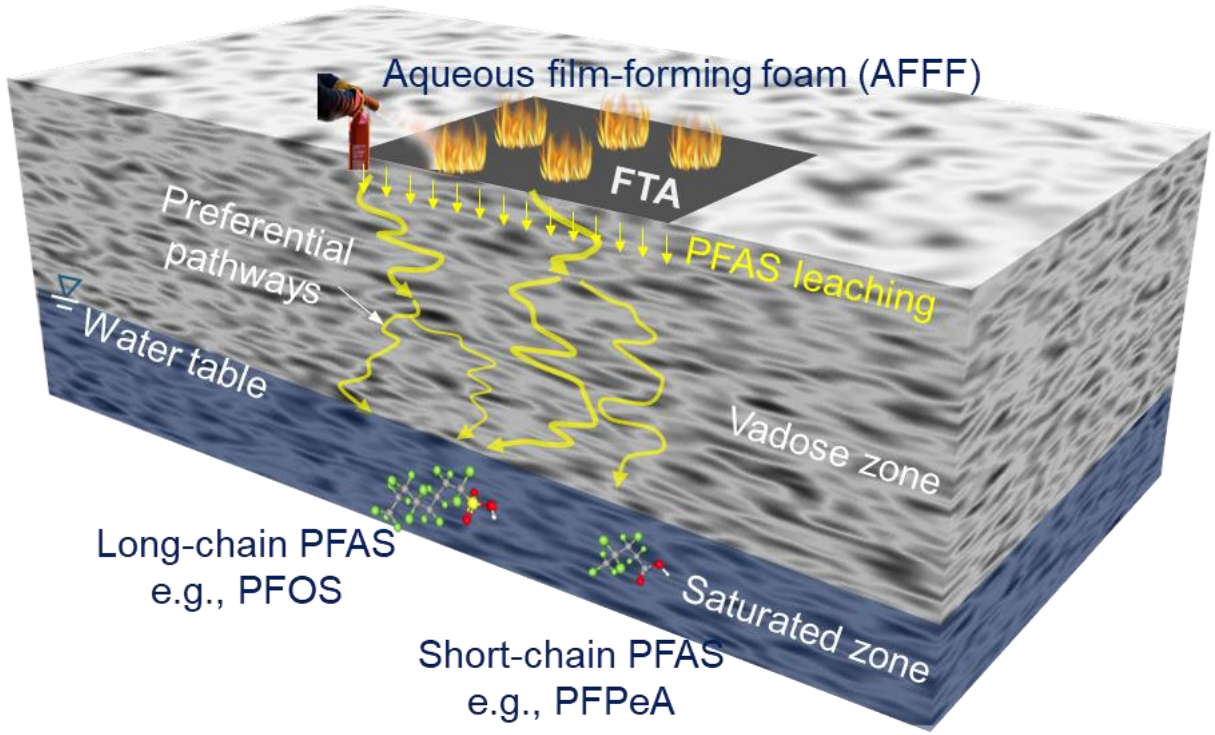


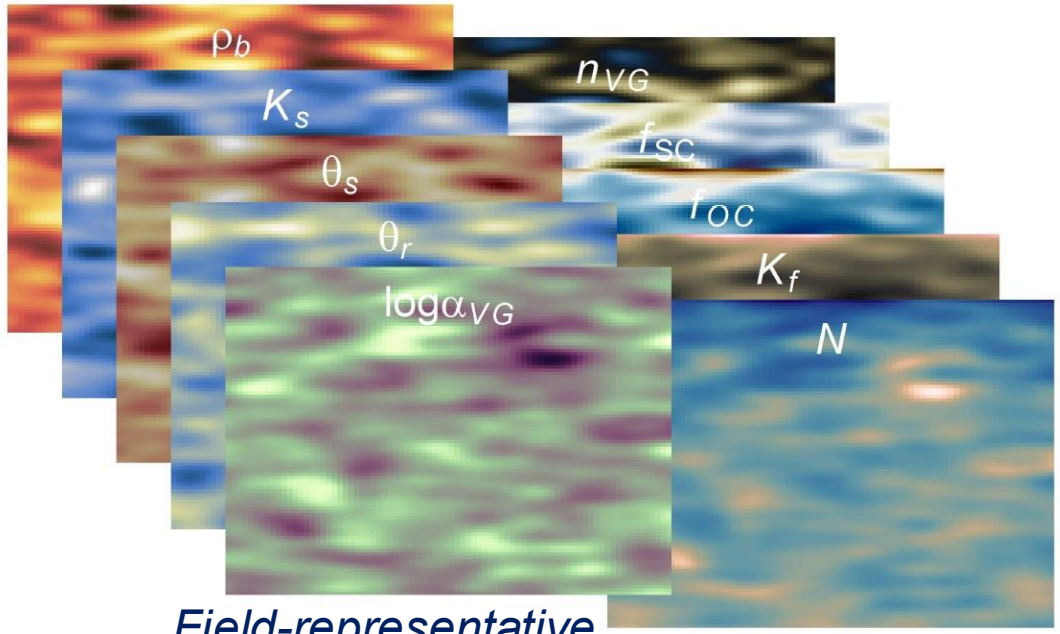
Accelerated leaching caused by heterogeneity

- PFAS leaching is accelerated in the presence of the preferential flow pathways.
- Along these flow pathways, air-water interfaces are destructed, which further accelerates PFAS migration.
- This is a phenomenon **unique** for PFAS, especially those long-chain compounds.

Zeng & Guo. 2021 AWR

More complex heterogeneities





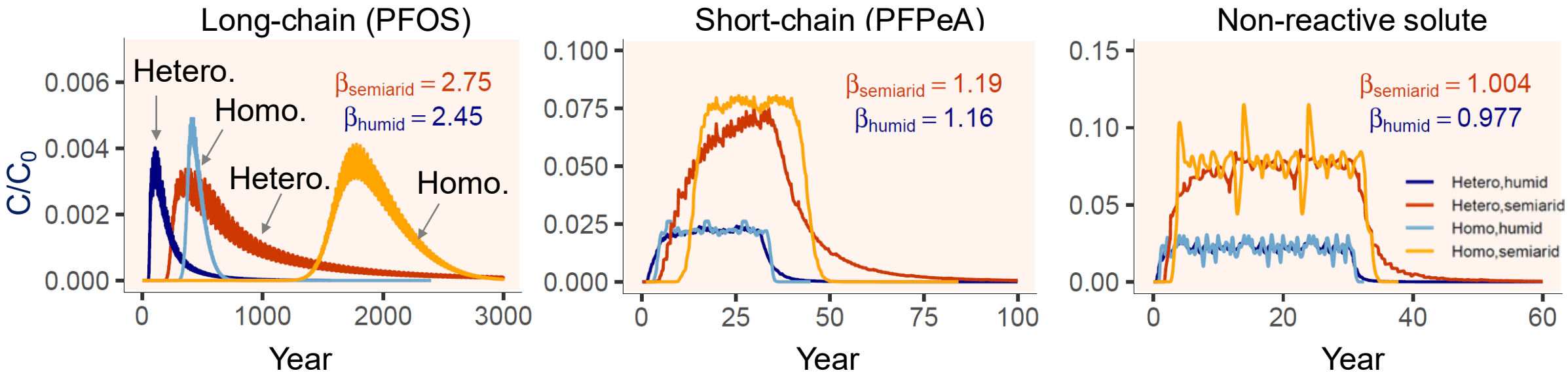
Field-representative heterogeneous fields

- Heterogeneous parameter fields are stochastically generated based on field measurements.
- Geochemical properties correlate with hydraulic parameters.

Zeng & Guo. 2023 GRL

Preferential flow uniquely accelerated PFAS leaching in soil

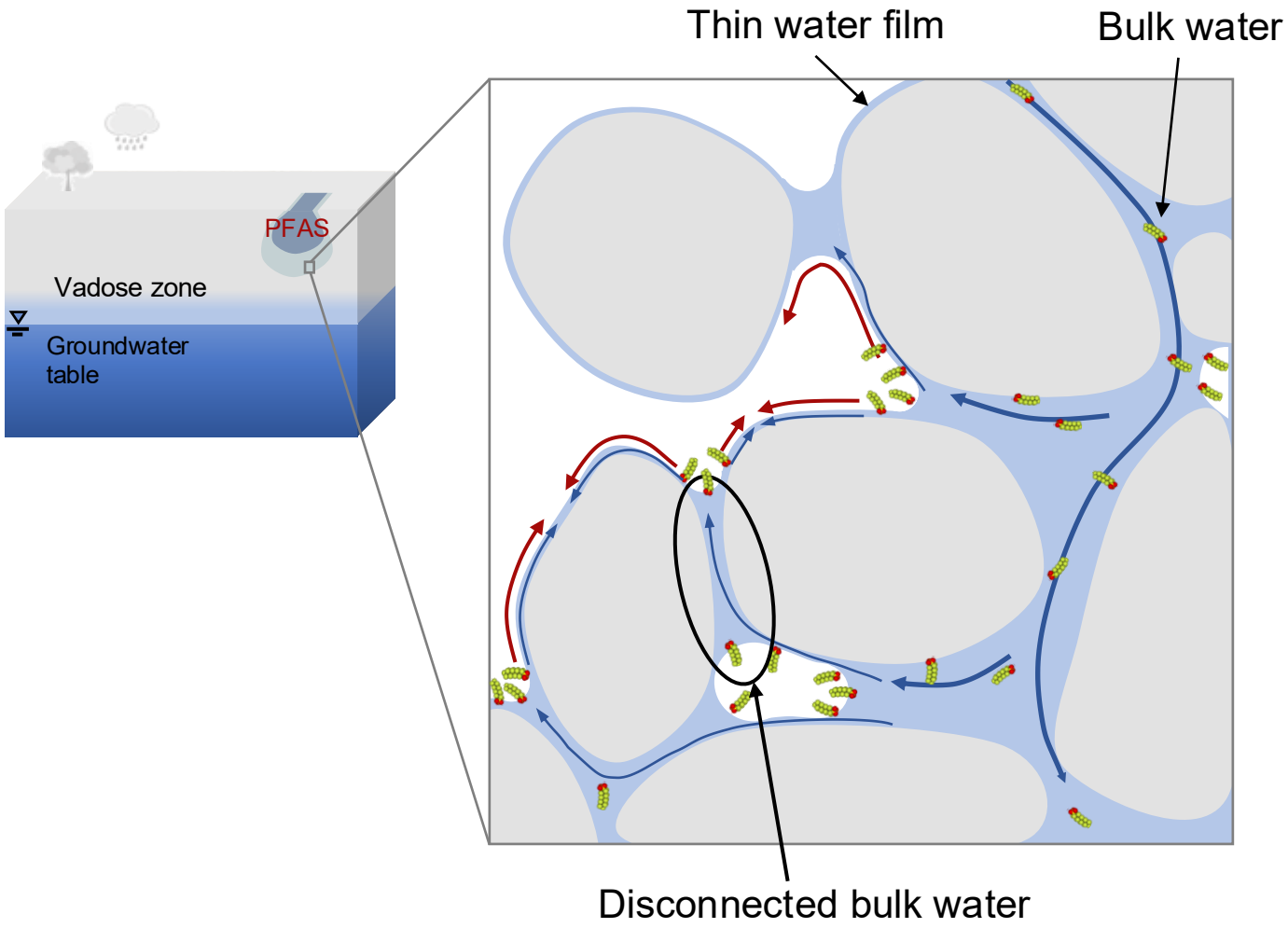
Breakthrough concentration (C/C_0)



- Accelerated movement of the center of mass through the vadose zone (β denotes acceleration ratio)
- The acceleration increases for:
  - Longer-chain PFAS
  - Vadose zones with stronger heterogeneities

Zeng & Guo. 2023 GRL

Air–water interfaces in thin water films vs. bulk water



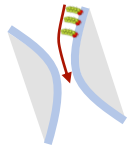
Are air–water interfaces in thin water films accessible?



- Connected bulk water
  Access bulk air–water interfaces via advection & diffusion in bulk water



- Disconnected bulk water
  Access air–water interfaces via diffusion in thin water films *(decreases with film thickness)*

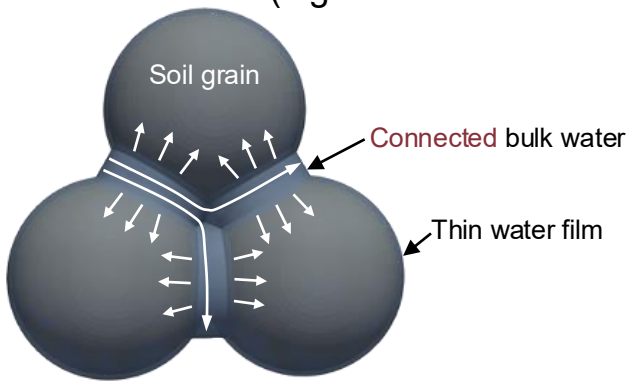


- Diffusion along air–water interfaces
  Access air–water interfaces via surface diffusion *(independent of film thickness)*

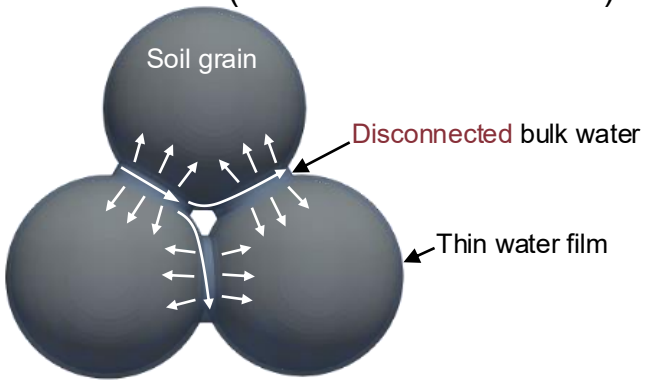
Chen & Guo. 2023 WRR

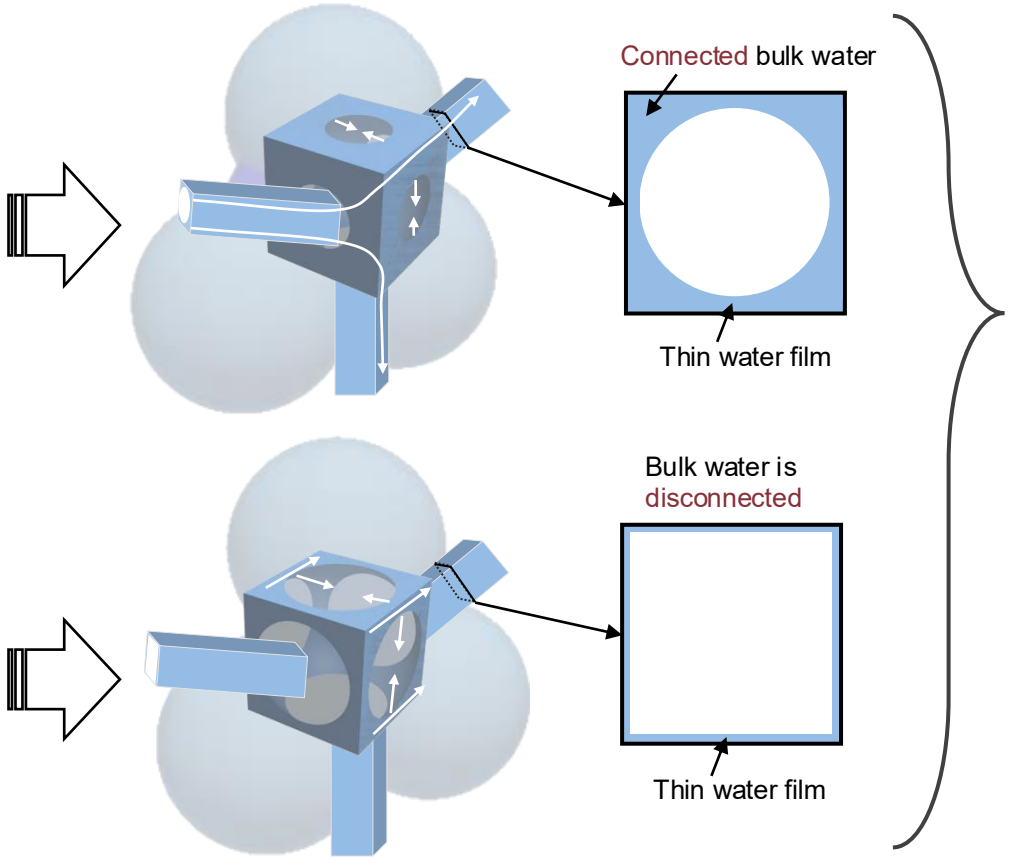
Pore-network modeling: water configuration, flow, & PFAS transport

- Wetter soil (higher water saturation)

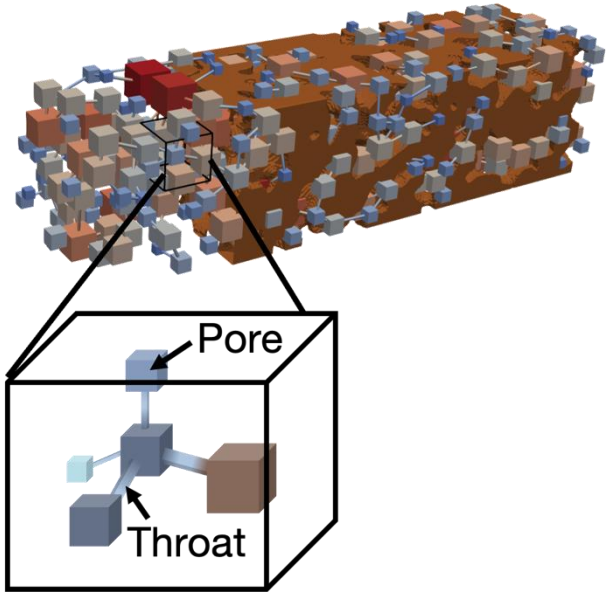


- Drier soil (lower water saturation)



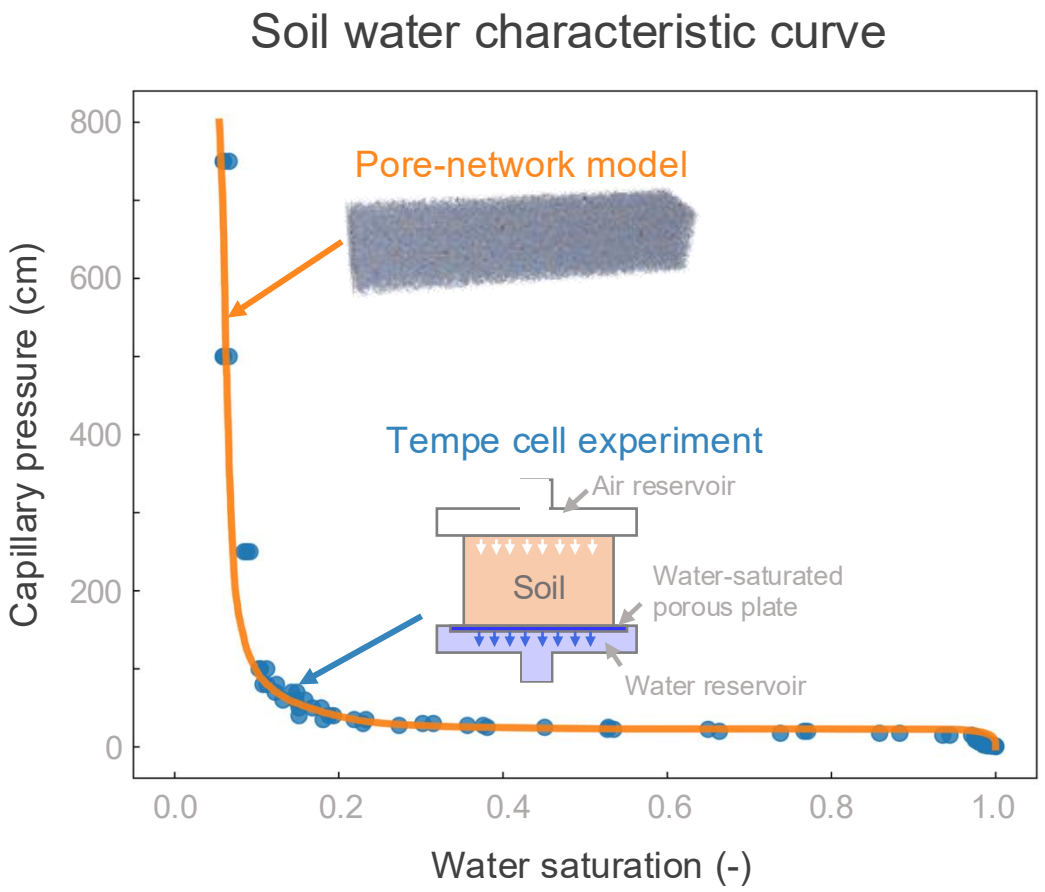


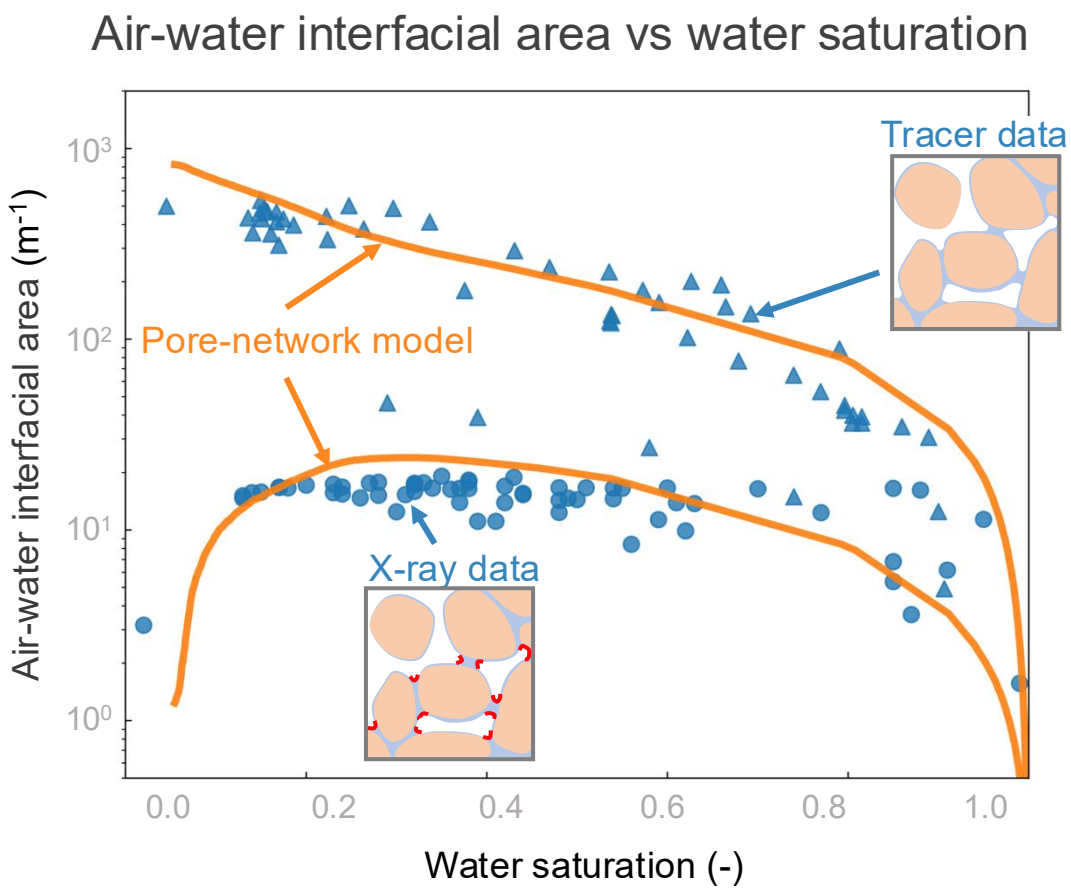
Represent water flow and PFAS transport in soil pore networks



Chen & Guo. 2023 WRR

Constraining the pore-network model with datasets for sandy soil

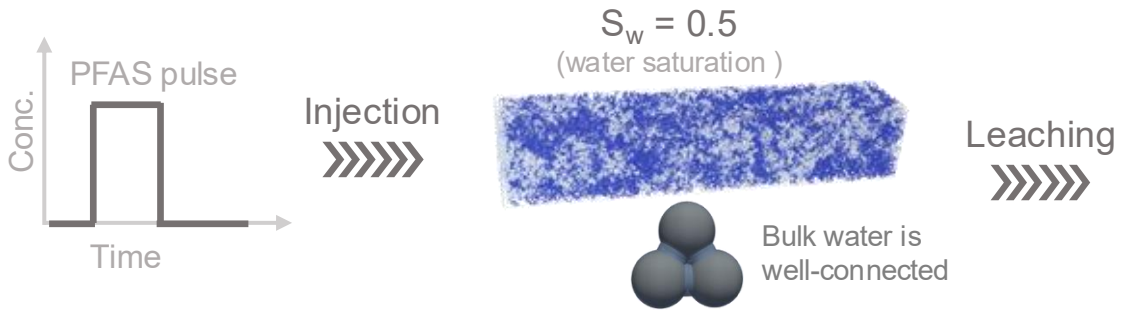


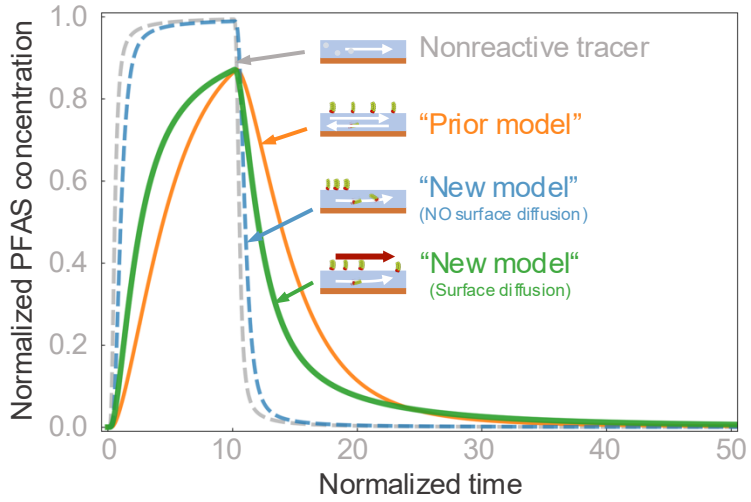


Chen & Guo (2023) WRR

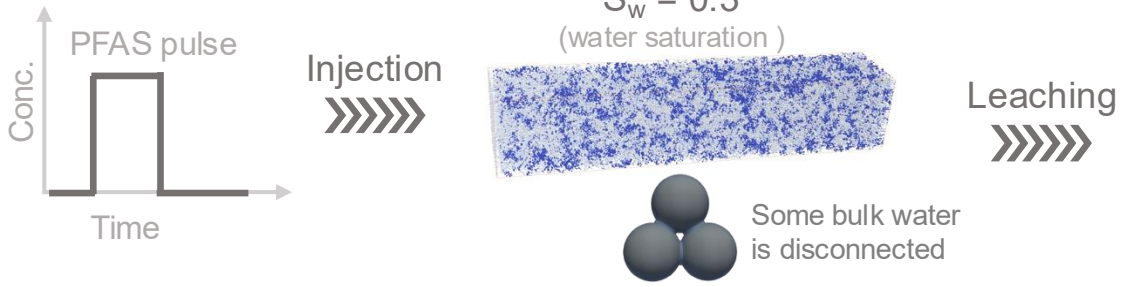
Modeling PFAS transport under wetter vs. drier conditions

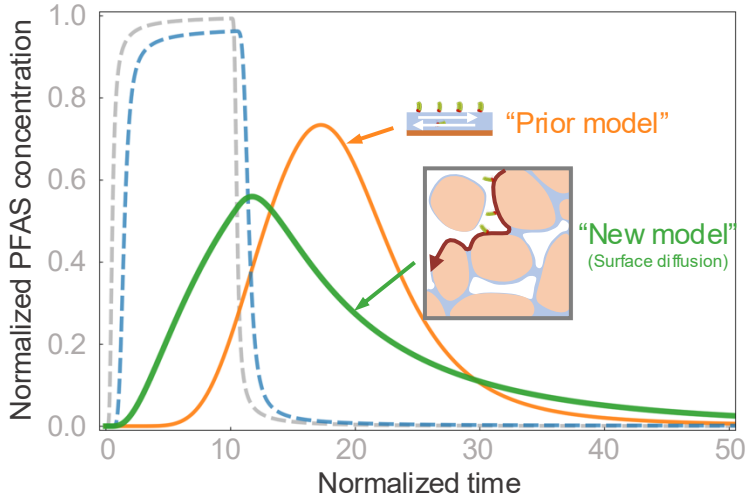
- Wetter condition

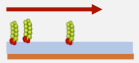
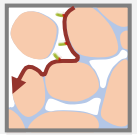




- Drier condition





Key processes for PFAS transport: 1. Diffusion along air–water interfaces  2. Slow transport in thin water film clusters 

Chen & Guo (2023) WRR

Can we develop simplified models for practical screening-type analysis?

PFAS transport in the vadose zone: simplified mathematical model

- Steady-state unsaturated flow:

$$q = K$$

- PFAS transport:

$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \boldsymbol{v} C) - \nabla \cdot (\theta \boldsymbol{D} \nabla C) = 0$$

Advection Dispersion

Linear and rate-limited solid-phase adsorption

$$C_{s,1} = F_s K_d C \qquad \frac{dC_{s,2}}{dt} = \alpha_s [(1 - F_s) K_d C - C_{s,2}]$$

Instantaneous Kinetic

Linear and equilibrium air-water interfacial adsorption

$$C_{aw} = A_{aw} K_{aw} C$$

Instantaneous

Guo, Zeng, Brusseau, Zhang. 2022 AWR

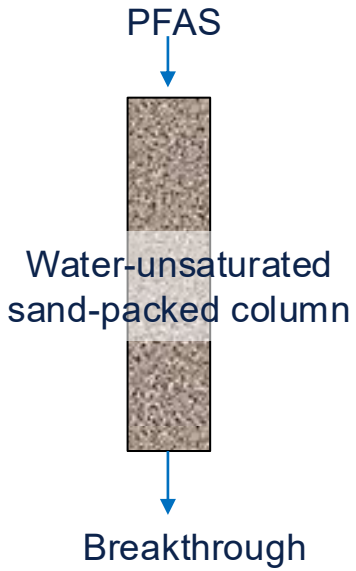
Generalization to **structured heterogeneous** soils:

- Dual-porosity, dual-permeability, and triple-porosity formulations
- Multi-site rate-limited
  - ✓ solid-phase sorption
  - ✓ air-water interfacial adsorption

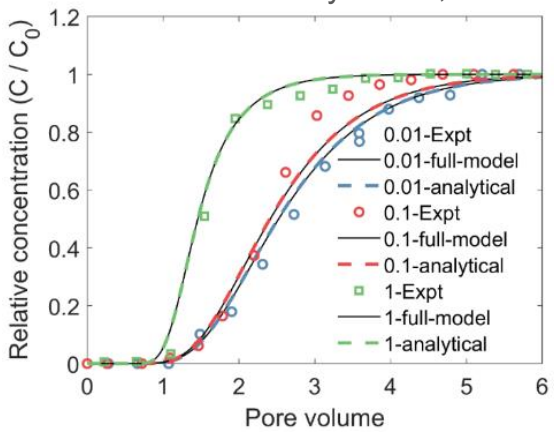
Chen & Guo. 2025 AWR

Code available on GitHub: https://github.com/GuoSFPLab/SemiAnalyticalSoln-PFAS-HeteroVZ

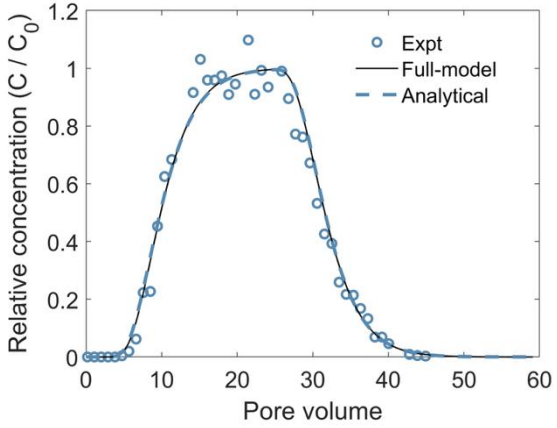
Model validation: vs. miscible-displacement experiments



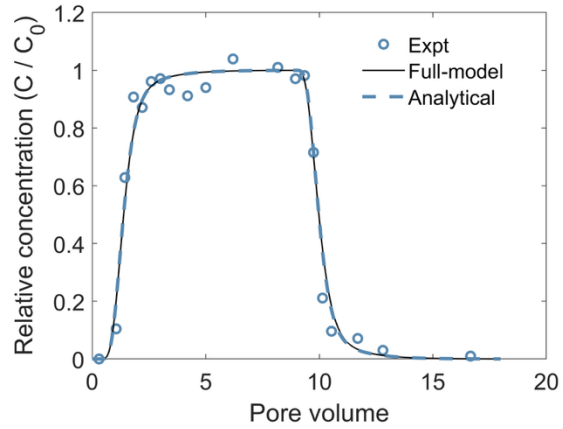
Experimental data set #1
PFOA (1 mg/L, 0.1 mg/L, 0.01 mg/L)
Measured data: Lyu et al., 2018



Experimental data set #2
PFOS (0.1 mg/L)
Measured data : Brusseau et al., 2021



Experimental data set #3
GenX (10 mg/L)
Measured data : Yan et al., 2021

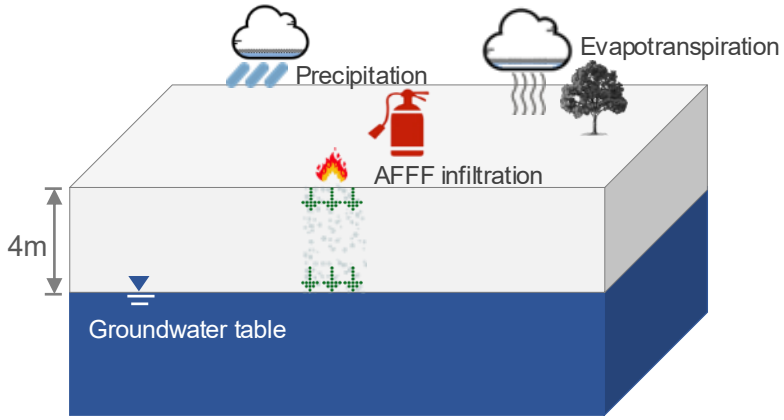


- Analytical solutions are identical to the numerical solutions of the full-process model.
- Independent model predictions match well with experimental data.

Guo, Zeng, Brusseau, Zhang. 2022 AWR

Model validation: vs. full-process model

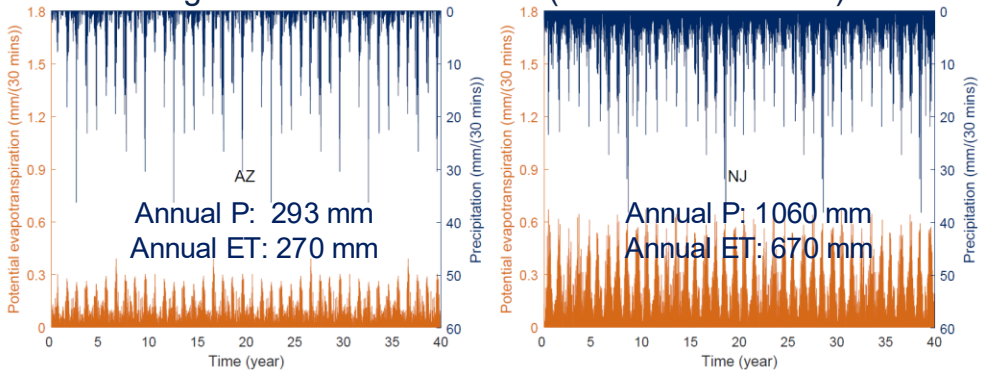
- PFAS contamination scenario at an AFFF-impacted fire training area site



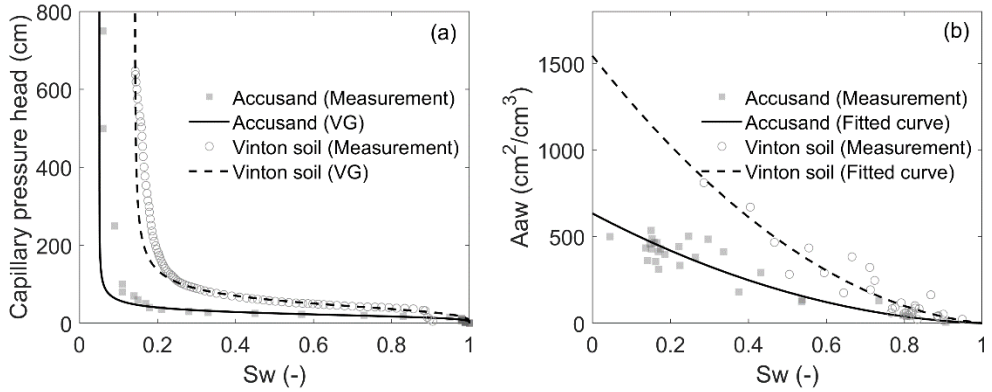
- ✓ Fire training: one session every 10 days lasting for 30 years
- ✓ Representative PFAS mixture in 1% diluted AFFF solution

  PFPeA: 0.23 mg/L, PFOA: 0.9 mg/L, PFHxS: 7.1 mg/L, PFOS: 100 mg/L

High resolution rainfall data (semi-arid vs. humid)

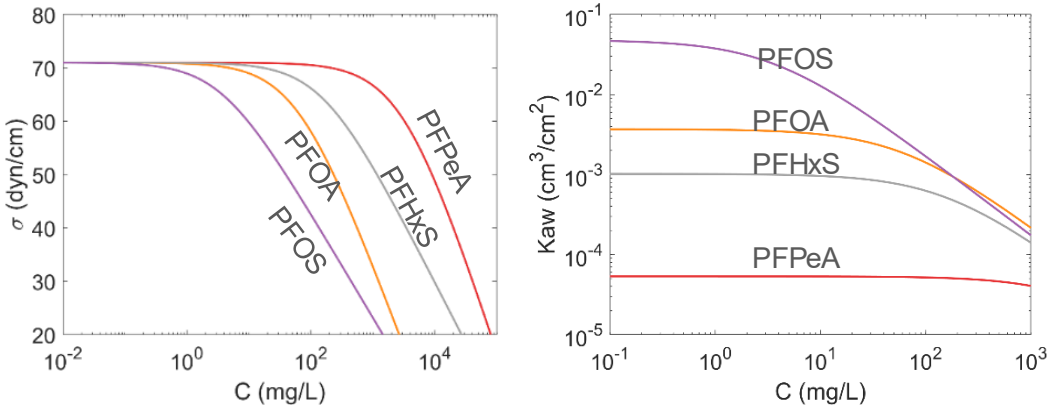


Two soil types (Accusand vs. Vinton)



Soil water characteristics | Air-water interfacial area

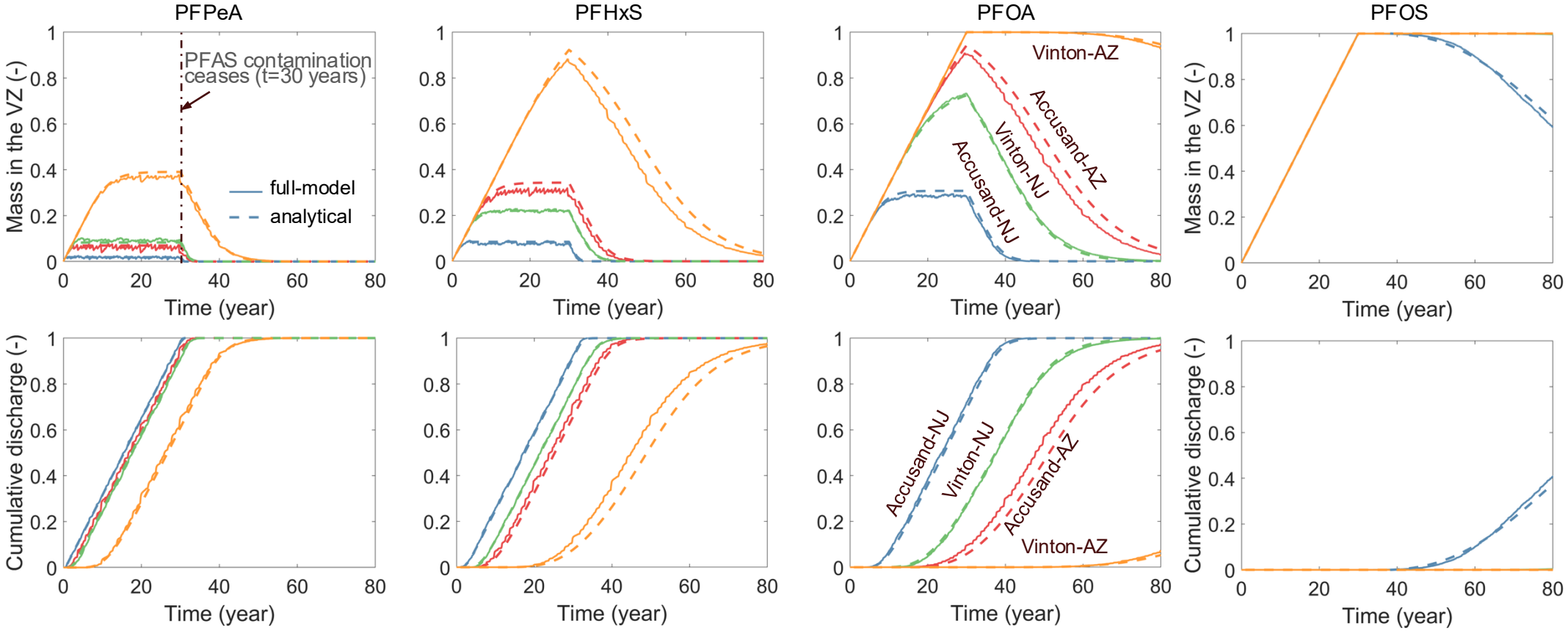
Four representative PFAS



Surface tension data | Air-water interfacial adsorption coeff.

Simulated long-term PFAS leaching in the vadose zone

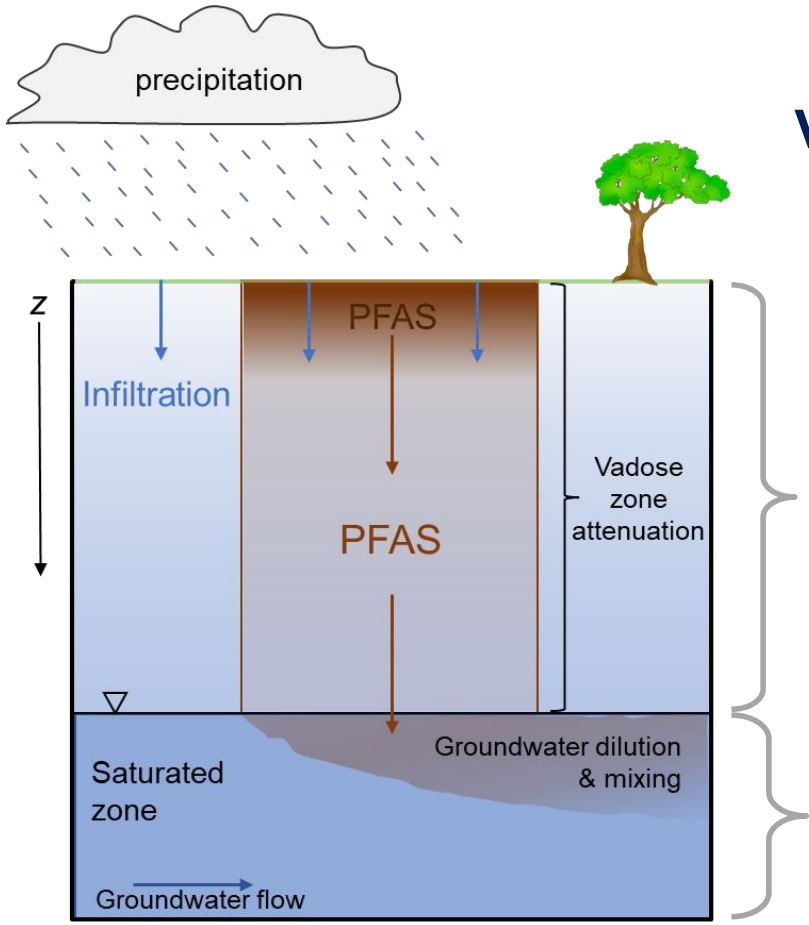
- PFAS retention in the vadose zone and mass discharge to groundwater



➢ Analytical model matches remarkably well with the full-process model for all four PFAS.

Guo, Zeng, Brusseau, Zhang. 2022 AWR

Derive site-specific soil screening levels (SSLs) for PFAS



Conceptual Model

Vadose zone (analytical **process-based** model, Guo et al. 2022)

- Assuming steady-state infiltration
- Transient transport
  - advection-dispersion
  - kinetic linear solid-phase adsorption
  - linear air-water interfacial adsorption

Representing VZ attenuation directly

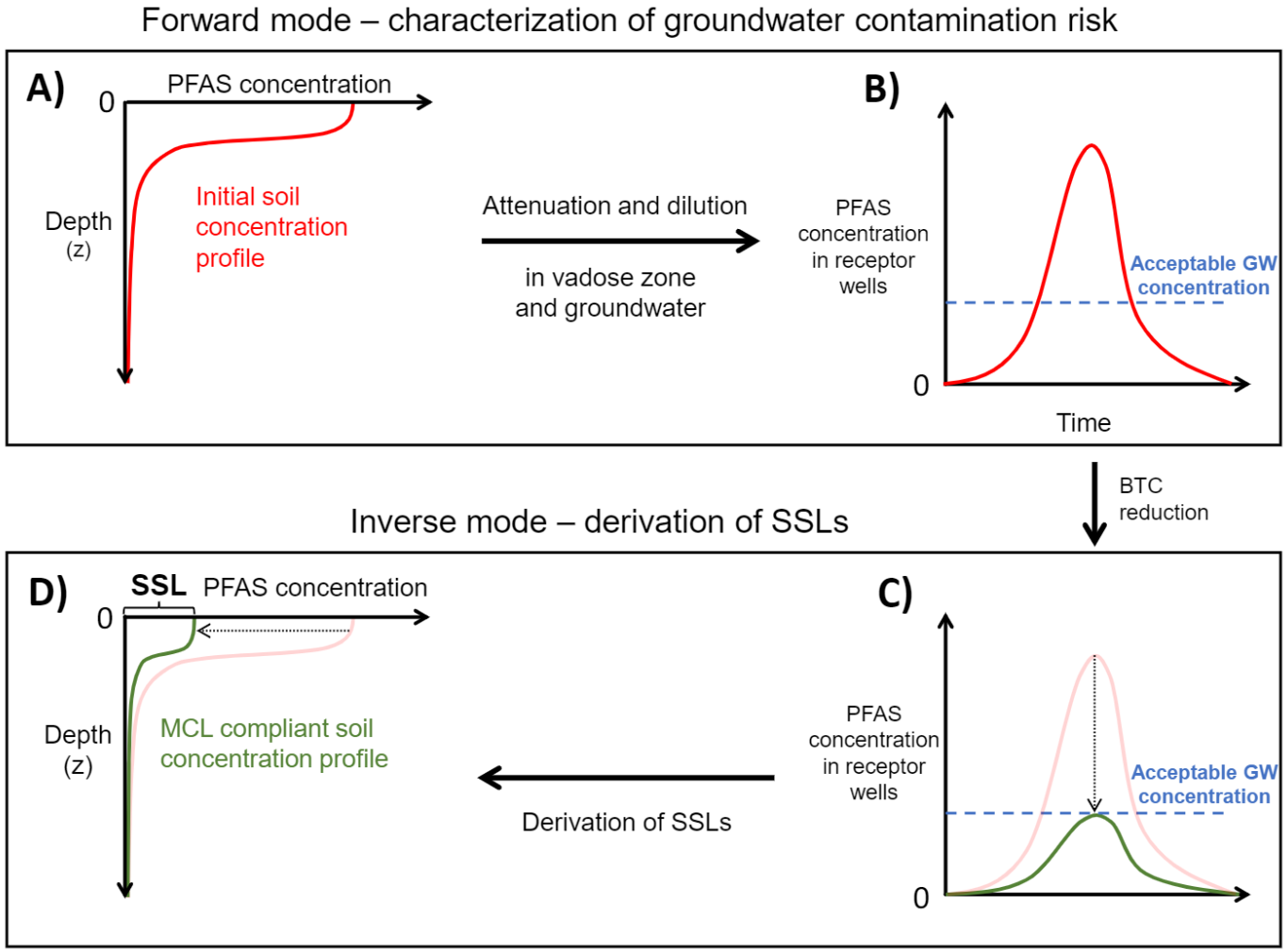
Transient mass discharge rates

Groundwater (GW mixing model, EPA 1996)

- Well-mixed between source zone and receptor point.
- Only accounts for dilution, not attenuation in groundwater.

Smith, Brusseau, Guo. 2024 WR

Derive site-specific soil screening levels (SSLs) for PFAS



- **Forward** mode evaluates groundwater contamination risk for a receptor point at a given contamination site.
- **Inverse** mode derives an SSL for a specific site that is protective of groundwater.

Smith, Brusseau, Guo. 2024 WR

Derived SSLs for example soils and PFAS in the vadose zone



- Standard EPA models not accounting for PFAS-specific interfacially-active properties can substantially overestimate PFAS leaching.
- SSLs derived by the new PFAS-specific model can be **1-2 orders of magnitudes greater**.

Smith, Brusseau, Guo. 2024 WR

Official Release of PFAS-LEACH Analytical Tool V1.0

Bo Guo · You
Associate Professor at University of Arizona
2w · Edited ·

What's included in v1.0?

The Excel tool includes the Tiers 3 and 4 models of PFAS-LEACH. Beta version for the Tiers 1 and 2 models will be released in the next six months.

PFAS-LEACH-Analytical (Tier 3): A set of analytical solutions that solve the partial differential equations for the leaching of PFAS in the vadose zone, and for deriving site-specific soil screening levels (SSLs).

PFAS-LEACH-DAF (Tier 4): A revision to the widely used dilution attenuation factor (DAF) approach in the USEPA SSL framework for determining site-specific SSLs for PFAS.

GuoSFPLab/PFAS-LEACH-Tier-3-4

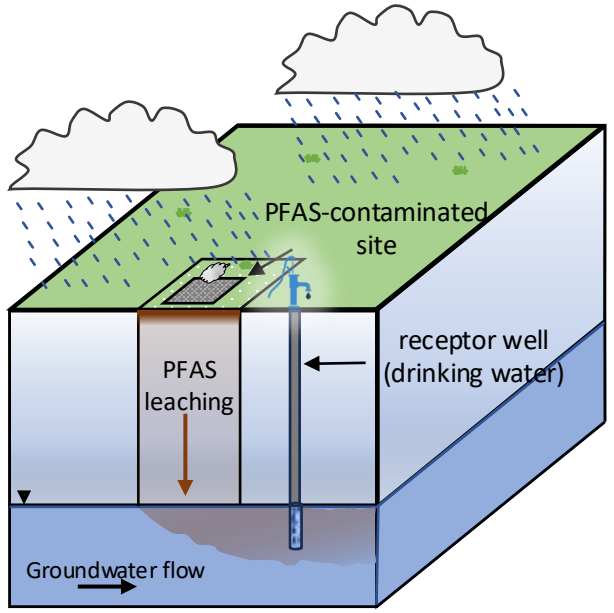
GitHub - GuoSFPLab/PFAS-LEACH-Tier-3-4: The Excel tool includes implementations of the Tiers 3...
github.com

Majid hassanizadeh and 250 others 23 comments · 15 reposts

Air Force Report (DRAFT, Dec 2025)

"Best Practices for Conducting PFAS Soil-to-Groundwater Leaching Evaluations"

PFAS-LEACH Excel tool is being recommended as the standard modeling tool



Module0

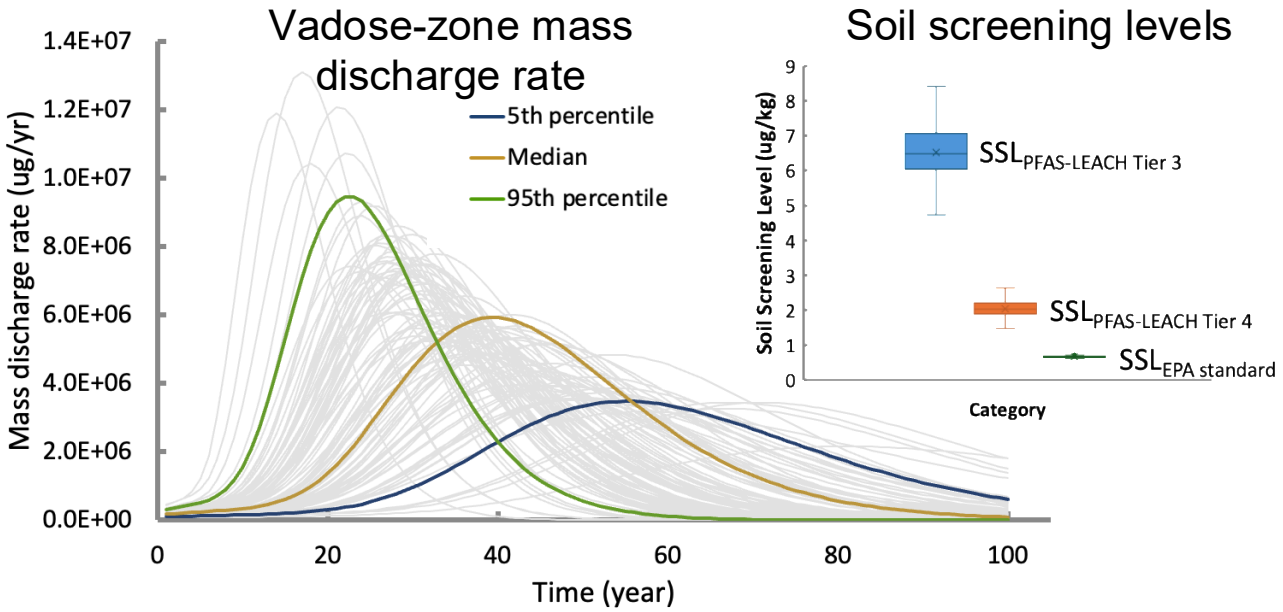
The basic module

Module1

Additional capability of parameter sensitivity analysis.

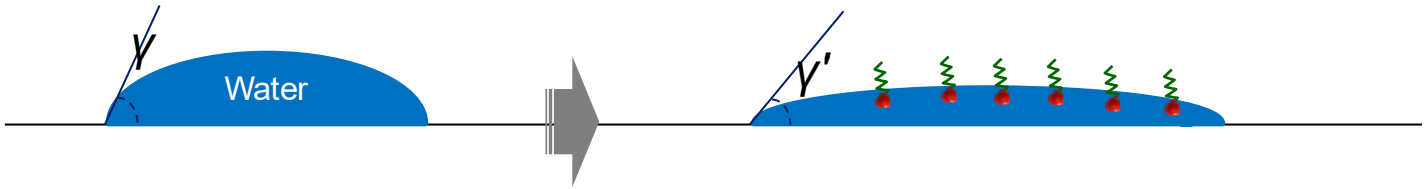
Module2

Additional capability of Monte-Carlo-type uncertainty analysis

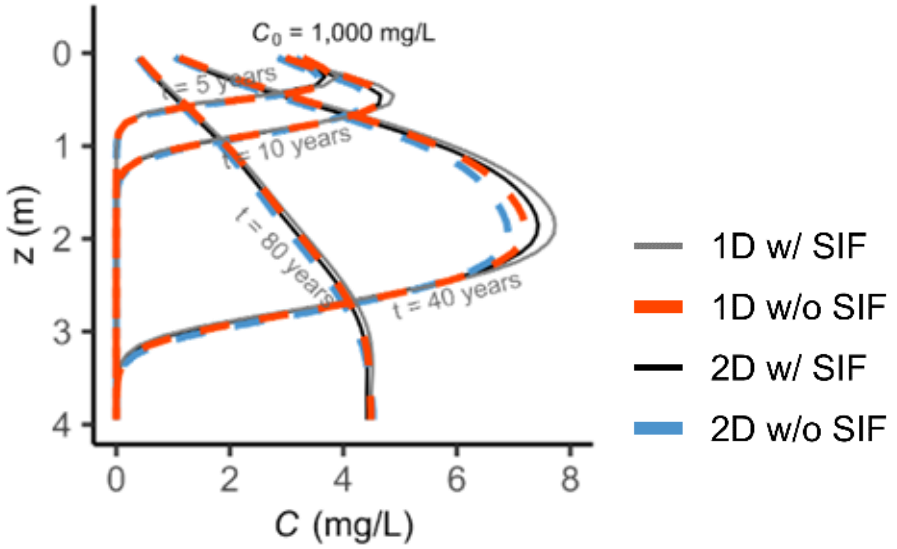


Additional processes: Surfactant-induced flow

Reduction in surface tension & contact angle can induce drainage



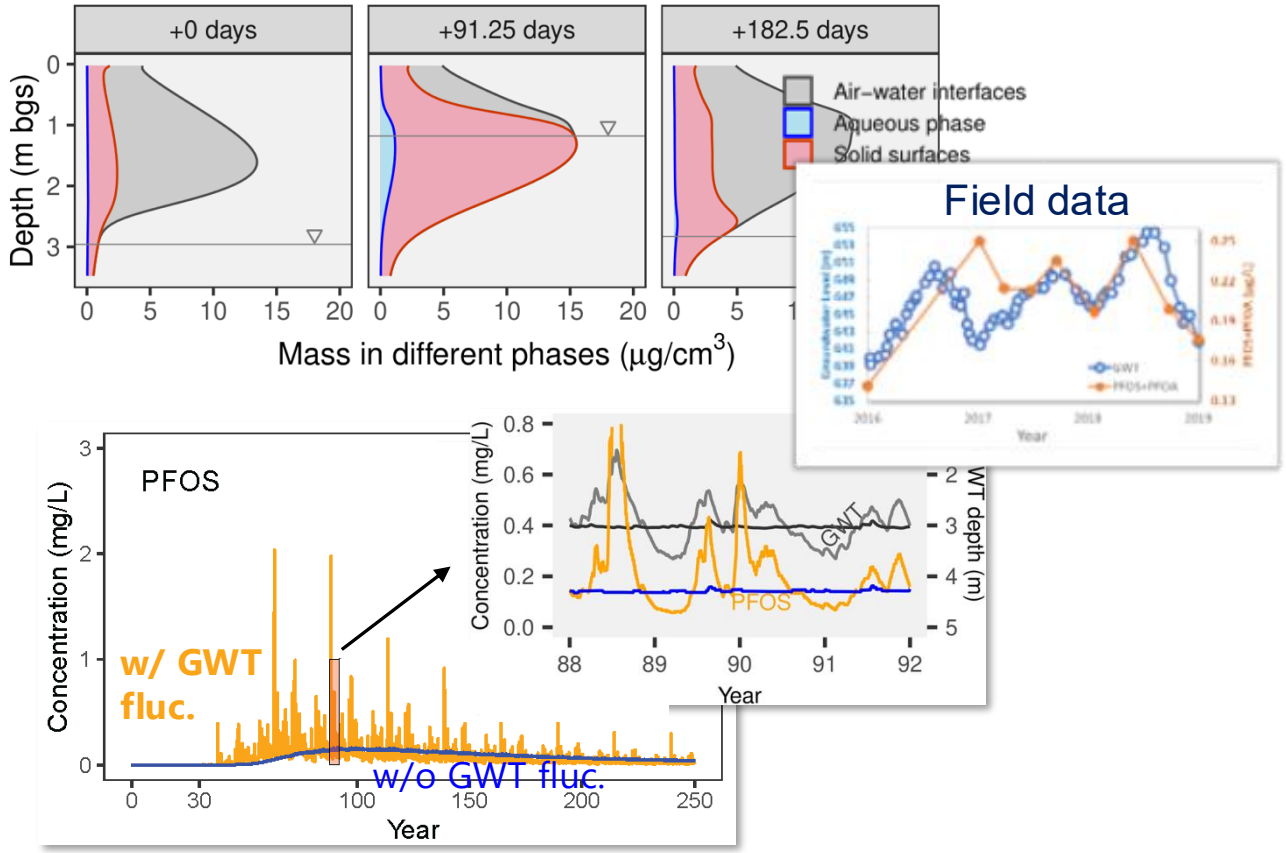
PFOS concentration profiles

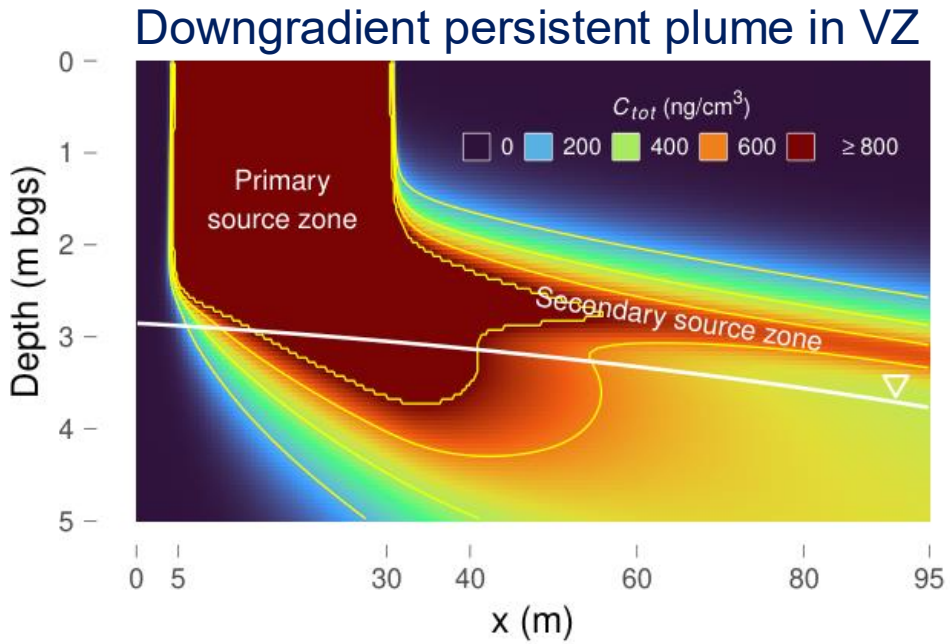


- Surfactant-induced flow modifies **local** air-water flow and PFAS transport, but has a minimum impact on the **long-term** leaching of PFAS in the vadose zone at many legacy sites.

Zeng, J. and Guo, B., 2021. Multidimensional simulation of PFAS transport and leaching in the vadose zone: Impact of surfactant-induced flow and subsurface heterogeneities. *Advances in Water Resources*, *155*, p.104015.

Additional processes: Groundwater table (GWT) fluctuation





- GWT rise and fall ➡ Enhanced source zone leaching and strong temporal variation of GW concentration.
- Lateral migration and GWT fluctuation ➡ A persistent plume in the downgradient vadose zone.

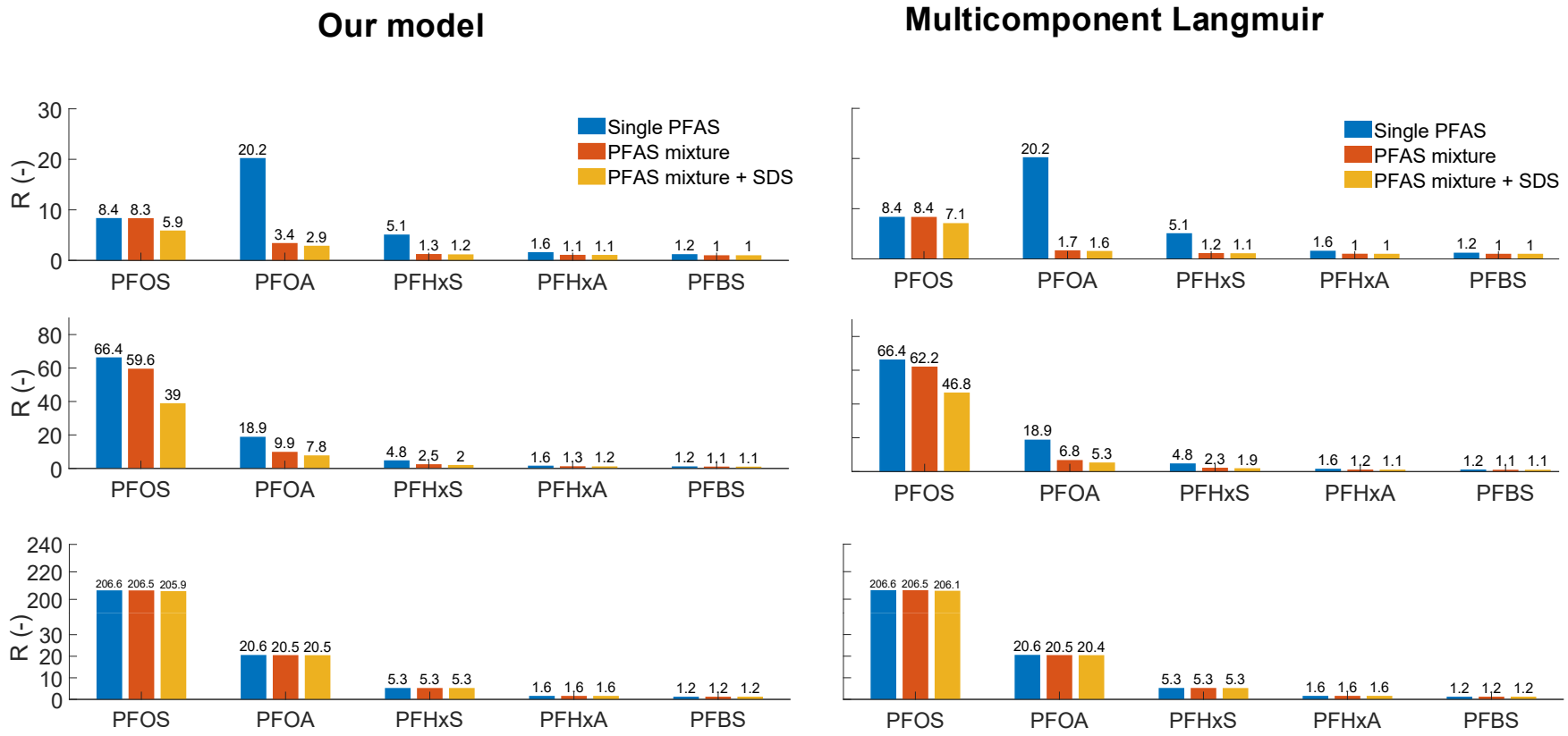
Zeng, J., Brusseau, M.L. and Guo, B., 2024. Modeling PFAS subsurface transport in the presence of groundwater table fluctuations: The impact on source-zone leaching and exploration of model simplifications. *Water Resources Research*

Additional processes: PFAS mixtures and hydrocarbon surfactants

Scenario #1
1% diluted AFFF solution

Scenario #2
Lysimeter-sampled porewater concentration from FTA site 1

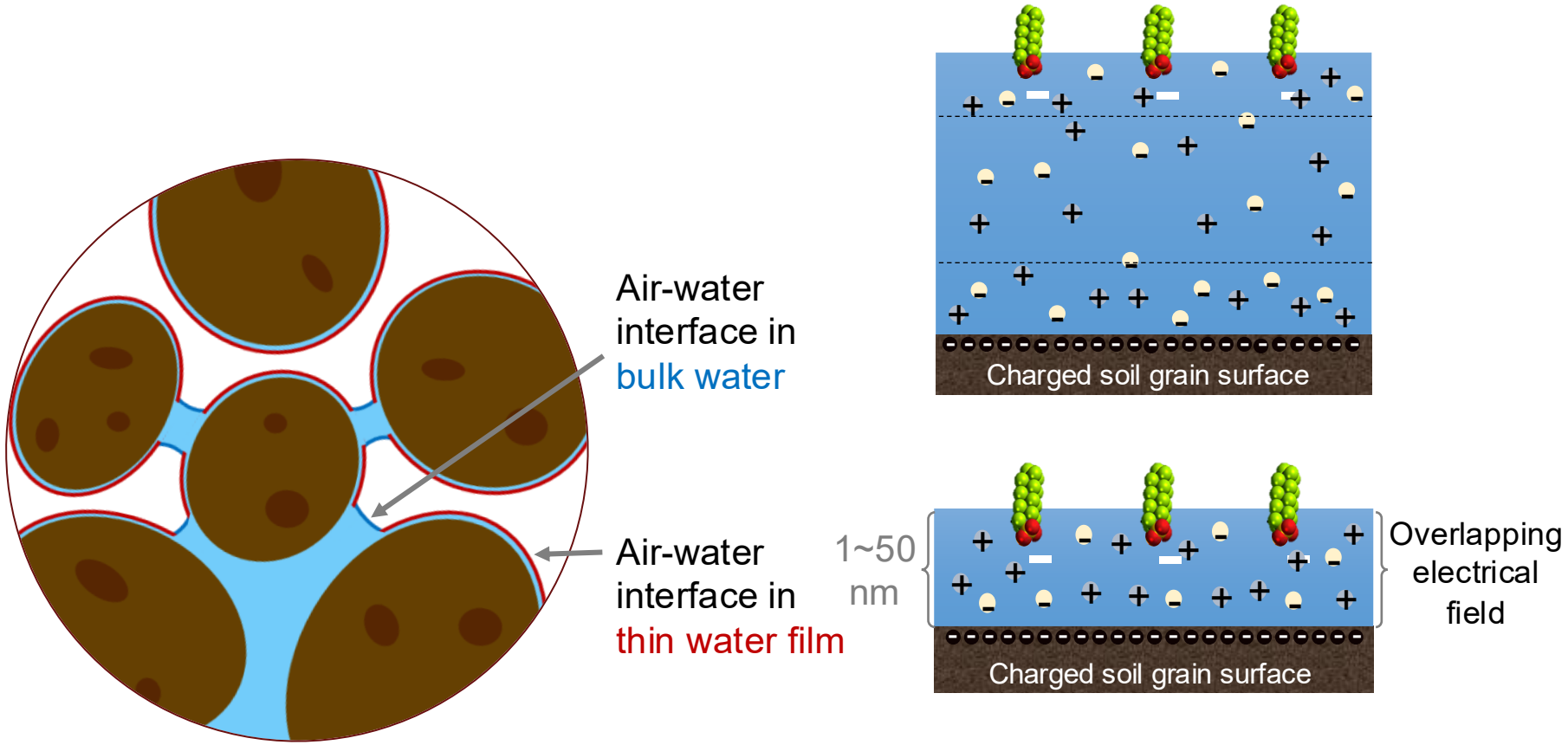
Scenario #3
Lysimeter-sampled porewater concentration from FTA site 2



- Competitive adsorption can significantly reduce PFAS retention in highly-contaminated sites, but likely not in lower concentration secondary contamination sites.
- The commonly used multicomponent Langmuir isotherm introduces errors due to model assumptions.

Guo, B., Saleem, H. and Brusseau, M.L., 2023. Predicting Interfacial Tension and Adsorption at Fluid−Fluid Interfaces for Mixtures of PFAS and/or Hydrocarbon Surfactants. *Environmental Science & Technology.*

Additional processes: Anomalous adsorption in thin-water-films



- The negatively charged soil grain surface **repels** the adsorbed anionic PFAS and significantly **reduces** (up to approximately 80%) adsorption at the thin-water-film air-water interfaces.

Zhang, W. and Guo, B., 2024. Anomalous adsorption of PFAS at the thin-water-film air-water interface in water-unsaturated porous media. *Water Resources Research*.

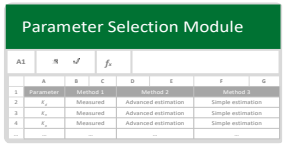
PFAS-LEACH – A Comprehensive Decision Support Platform for Predicting PFAS Leaching in Source Zones

Computational Cost & Input Parameters

Model Complexity

i. Four models spanning a wide range of complexity
ii. Comprehensive parameter selection module
iii. Documentation and user manual

Utility Toolbox



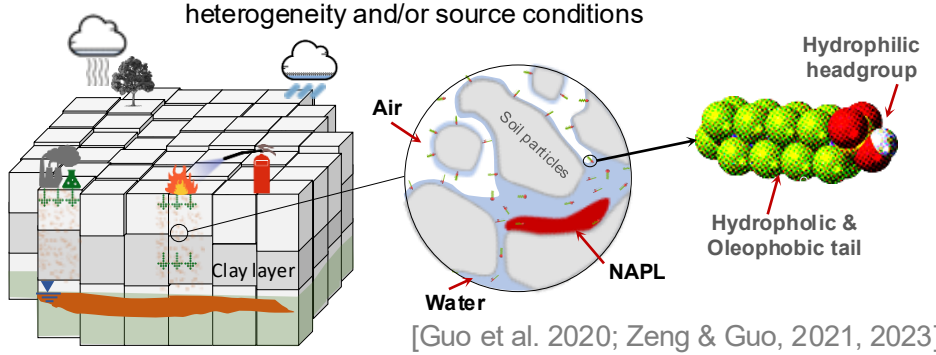
Applications:

- Simulate retention & transport in multiphase systems
- Quantify mass discharge to groundwater
- Support risk assessments for soil-groundwater route
- Determine soil screening levels
- Determine remedial action targets
- Evaluate remedial action performance

Tier 1 – PFAS-LEACH-COMP

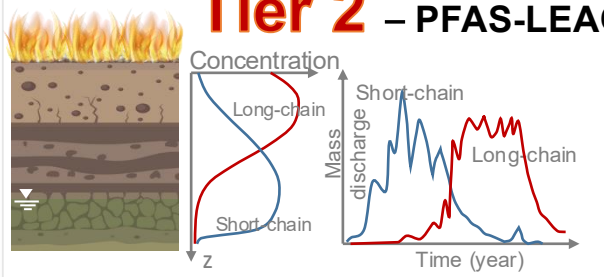
Features: Full-process model in 3D

- *Anticipated application:* Sites with sufficient data; complex spatial heterogeneity and/or source conditions



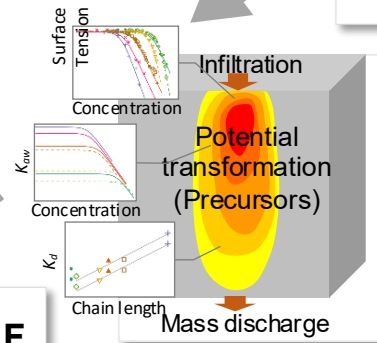
[Guo et al. 2020; Zeng & Guo, 2021, 2023]

Tier 2 – PFAS-LEACH-Numerical-1D



Features: 1D numerical solution
Anticipated application: Insignificant lateral heterogeneity; uncertainty quantification

[Zeng et al. 2021]

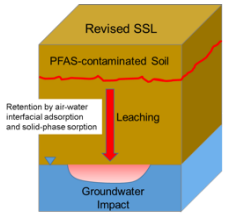


Tier 3 – PFAS-LEACH-Analytical

Excel

Features: 1D analytical solutions Implemented in Excel
Anticipated application: Limited data; early stage of site management; order-of-magnitude estimate.

[Guo et al. 2022; Smith et al., 2024]



Tier 4 – PFAS-LEACH-DAF

Features: Revised EPA dilution-attenuation model

- Simplified mass-balance calculations

Anticipated application: Initial screening of sites

[Brusseau & Guo, 2023]

❖ **PFAS-LEACH Tiers 2-4 have been released:**
https://github.com/GuoSFPLab

Testing PFAS-LEACH models using pilot-scale experiments & field data





ESTCP Project ER21-5041
PI: Bo Guo

Excavating the soil cores and monolith from the Davis Monthan AFB (Tucson, AZ).





Soil cores and monolith transported to the University of Arizona for leaching experiments.

Take-home message

- Mathematical models with varying complexity representing PFAS-specific transport processes.
- Air-water interfacial adsorption leads to strong retention of (long-chain) PFAS.
- Preferential flow destroys air-water interfaces and accelerates PFAS leaching to deep vadose zone.
- The simplified model provides an efficient and accurate screening-type tool for quantifying vadose-zone PFAS leaching.
- Surfactant-induced flow appears to have a minor impact on long-term PFAS leaching.
- Additional processes to consider: GWT fluctuation, Multicomponent competitive adsorption, Anomalous adsorption in thin-water-film air-water interfaces.

Project ER21-5041
EAR 2023351
EAR 2054575
CAREER 2237015